A Living Review of Quantum Information Science in High Energy Physics

Pamela Pajarillo, So Chigusa, Sokratis Trifinopoulos, Jesse Thaler

Center for Theoretical Physics - a Leinweber Institute, Massachusetts Institute of Technology Cambridge, MA 02139, USA

E-mail: pampaja@mit.edu, schigusa@mit.edu, trifinos@mit.edu, jthaler@mit.edu

Abstract: Inspired by "A Living Review of Machine Learning for Particle Physics"[1], the goal of this document is to provide a nearly comprehensive list of citations for those developing and applying quantum information approaches to experimental, phenomenological, or theoretical analyses. Applications of quantum information science to high energy physics is a relatively new field of research. As a living document, it will be updated as often as possible with the relevant literature with the latest developments. Suggestions are most welcome.

[1]See https://github.com/iml-wg/HEPML-LivingReview.

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1 Introduction

The purpose of this note is to collect references for quantum information science as applied to particle and nuclear physics. The papers are listed in reverse chronological order. In order to be as useful as possible, this document will continually change. Please check back [2] regularly. You can simply download the .bib file to get all of the latest references. Suggestions are most welcome.

2 Nuclear and Particle Physics in Quantum Information Science

2.1 Reviews and Whitepapers and Proceedings

2.1.1 Reviews

2.1.1.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero

[2] See https://github.com/PamelaPajarillo/NUPAQIS-LivingReview.

- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

2.1.1.2 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

2.1.1.3 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.1.1.4 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.1.1.5 Review on Quantum Computing for Lattice Field Theory [5]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in PoS:** 01 February 2023

2.1.1.6 Machine learning for anomaly detection in particle physics [6]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klaeboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

2.1.1.7 Quantum entanglement and Bell inequality violation at colliders [7]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

2.1.2 Whitepapers and Proceedings

2.1.2.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

2.1.2.2 Mechanical quantum sensing in the search for dark matter [9]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.1.2.3 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [10]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

2.1.2.4 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [11]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.1.2.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.1.2.6 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

2.1.2.7 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.1.2.8 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou
- **Posted on arXiv:** 28 March 2022

2.1.2.9 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

2.1.2.10 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022

- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.1.2.11 Quantum computing hardware for HEP algorithms and sensing [18]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti

- **Posted on arXiv:** 18 April 2022

2.1.2.12 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage

- **Posted on arXiv:** 14 September 2022

2.1.2.13 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023

- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.1.2.14 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik,

J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

2.2 Anomaly Detection

2.2.1 Continuous Variable Quantum Computing

2.2.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [22]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.2.2 Quantum Annealing

2.2.2.1 A Quantum Algorithm for Model-Independent Searches for New Physics [23]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

2.2.3 Quantum Machine Learning - Supervised Methods

2.2.3.1 Unravelling physics beyond the standard model with classical and quantum anomaly detection [24]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

2.2.4 Quantum Machine Learning - Unsupervised Methods

2.2.4.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [25]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

2.2.5 Quantum Machine Learning - Variational Quantum Algorithms

2.2.5.1 Anomaly detection in high-energy physics using a quantum autoencoder [26]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

2.2.5.2 Quantum anomaly detection for collider physics [27]

- **Authors:** Sulaiman Alvi, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 16 June 2022
- **Published in JHEP:** 22 February 2023

2.2.6 Crosslists

2.2.6.1 Machine learning for anomaly detection in particle physics [6]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klaeboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

2.3 Beyond the Standard Model

2.3.1 Quantum Algorithms - Grover's Search Algorithm

2.3.1.1 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [28]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

2.3.2 Quantum Entanglement and Bell Inequalities

2.3.2.1 Flavor patterns of fundamental particles from quantum entanglement? [29]

- **Authors:** Jesse Thaler, Sokratis Trifinopoulos
- **Posted on arXiv:** 30 October 2024
- **Published in Phys.Rev.D:** 01 March 2025

2.3.3 Quantum Machine Learning - Variational Quantum Algorithms

2.3.3.1 Hybrid Quantum-Classical Graph Convolutional Network [30]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

2.3.4 Crosslists

2.3.4.1 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

2.3.4.2 Event Classification with Quantum Machine Learning in High-Energy Physics [31]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.3.4.3 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [32]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.3.4.4 Quantum convolutional neural networks for high energy physics data analysis [33]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.3.4.5 Quantum algorithm for the classification of supersymmetric top quark events [34]

- **Authors:** Pedrame Bargassa, Timothèe Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothèe Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.3.4.6 Unravelling physics beyond the standard model with classical and quantum anomaly detection [24]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

2.3.4.7 Quantum anomaly detection in the latent space of proton collision events at the LHC [25]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

2.3.4.8 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik,

J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

2.3.4.9 Quantum Metric Learning for New Physics Searches at the LHC [35]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

2.4 Dark Matter - Particle-like Dark Matter

2.4.1 Quantum Algorithms - Quantum Simulations

2.4.1.1 Quantum simulations of dark sector showers [36]

- **Authors:** So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 26 April 2022
- **Published in Phys.Lett.B:** 26 September 2022

2.4.2 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.4.2.1 Trapped Electrons and Ions as Particle Detectors [37]

- **Authors:** Daniel Carney, Hartmut Häffner, David C. Moore, Jacob M. Taylor
- **Posted on arXiv:** 12 April 2021
- **Published in Phys Rev Lett, Vol. 127, No. 6, 2021. "Editor's suggestion":** 05 August 2021

2.4.2.2 Millicharged Dark Matter Detection with Ion Traps [38]

- **Authors:** Dmitry Budker, Peter W. Graham, Harikrishnan Ramani, Ferdinand Schmidt-Kaler, Christian Smorra, Stefan Ulmer
- **Posted on arXiv:** 11 August 2021
- **Published in PRX Quantum:** 01 February 2022

2.4.3 Quantum Sensors - Optomechanical Sensors

2.4.3.1 Search for Kilogram-scale Dark Matter with Precision Displacement Sensors [39]

- **Authors:** Akio Kawasaki
- **Posted on arXiv:** 26 August 2018
- **Published in Phys. Rev. D 99, 023005 (2019):** 07 January 2019

2.4.3.2 Proposal for gravitational direct detection of dark matter [40]

- **Authors:** Daniel Carney, Sohitri Ghosh, Gordan Krnjaic, Jacob M. Taylor
- **Posted on arXiv:** 01 March 2019
- **Published in Phys.Rev.D:** 14 October 2020

2.4.3.3 Search for composite dark matter with optically levitated sensors [41]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

2.4.3.4 Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors [42]

- **Authors:** Gadi Afek, Daniel Carney, David C. Moore
- **Posted on arXiv:** 05 November 2021
- **Published in Phys.Rev.Lett.:** 09 March 2022

2.4.3.5 Models of ultraheavy dark matter visible to macroscopic mechanical sensing arrays [43]

- **Authors:** Carlos Blanco, Bahaa Elshimy, Rafael F. Lang, Robert Orlando
- **Posted on arXiv:** 29 December 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.4.3.6 Requirements on quantum superpositions of macro-objects for sensing neutrinos [44]

- **Authors:** Eva Kilian, Marko Toroš, Frank F. Deppisch, Ruben Saakyan, Sougato Bose
- **Posted on arXiv:** 27 April 2022
- **Published in Phys. Rev. Research 5, 023012 (2023):** 07 April 2023

2.4.3.7 Searches for Massive Neutrinos with Mechanical Quantum Sensors [45]

- **Authors:** Daniel Carney, Kyle G. Leach, David C. Moore
- **Posted on arXiv:** 12 July 2022
- **Published in Phys.Rev.X:** 01 February 2023

2.4.3.8 Optomechanical dark matter instrument for direct detection [46]

- **Authors:** Christopher G. Baker, Warwick P. Bowen, Peter Cox, Matthew J. Dolan, Maxim Goryachev, Glen Harris
- **Posted on arXiv:** 16 June 2023
- **Published in Phys. Rev. D 110, 043005 (2024):** 02 August 2024

2.4.3.9 Probing levitodynamics with multi-stochastic forces and the simple applications on the dark matter detection in optical levitation experiment [47]

- **Authors:** Xi Cheng, Ji-Heng Guo, Wenyu Wang, Bin Zhu
- **Posted on arXiv:** 07 December 2023
- **Published in Nucl. Phys. B 1010 (2025) 116780:** 19 December 2024

2.4.3.10 Dark Matter Searches with Levitated Sensors [48]

- **Authors:** Eva Kilian, Markus Rademacher, Jonathan M.H. Gosling, Julian H. Iacoponi, Fiona Alder, Marko Toroš, Antonio Pontin, Chamkaur Ghag, Sougato Bose, Tania S. Monteiro, P.F. Barker
- **Posted on arXiv:** 31 January 2024
- **Published in AVS Quantum Sci. 6, 030503 (2024):** 2024

2.4.3.11 Probing light particles with optically trapped sensors through nucleon scattering [49]

- **Authors:** Bhaskar Dutta, Dilip Kumar Ghosh, Sk Jeesun
- **Posted on arXiv:** 31 January 2025
- **Published in Phys.Rev.D:** 01 January 2026

2.4.3.12 Mechanical sensors for ultraheavy dark matter searches via long-range forces [50]

- **Authors:** Juehang Qin, Dorian W.P. Amaral, Sunil A. Bhave, Erqian Cai, Daniel Carney, Rafael F. Lang, Shengchao Li, Alberto M. Marino, Giacomo Marocco, Claire Marvinney, Jared R. Newton, Jacob M. Taylor, Christopher Tunnell
- **Posted on arXiv:** 14 March 2025
- **Published in Phys.Rev.D 112 (2025) 7, 072003:** 01 October 2025

2.4.4 Quantum Sensors - Solid State Sensors

2.4.4.1 Dark matter direct detection with quantum dots [51]

- **Authors:** Carlos Blanco, Rouven Essig, Marivi Fernandez-Serra, Harikrishnan Ramani, Oren Slone
- **Posted on arXiv:** 10 August 2022
- **Published in Phys.Rev.D:** 01 May 2023

2.4.4.2 Dark Matter Induced Power in Quantum Devices [52]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 17 October 2022
- **Published in Phys. Rev. Lett. 132, 121801 (2024):** 22 March 2024

2.4.4.3 Transmon Qubit constraints on dark matter-nucleon scattering [53]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 30 April 2024
- **Published in JHEP:** 25 July 2024

2.4.4.4 Quantum parity detectors: A qubit-based particle-detection scheme with meV thresholds for rare-event searches [54]

- **Authors:** Karthik Ramanathan, Brandon J. Sandoval, John E. Parker, Lalit M. Joshi, Andrew D. Beyer, Pierre M. Echternach, Serge Rosenblum, Sunil R. Golwala
- **Posted on arXiv:** 27 May 2024
- **Published in APS Open Science:** 01 May 2026

2.4.5 Crosslists

2.4.5.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

2.4.5.2 Mechanical quantum sensing in the search for dark matter [9]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.4.5.3 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova

- **Posted on arXiv:** 14 March 2022

2.4.5.4 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

2.4.5.5 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

2.5 Dark Matter - Wave-like Dark Matter

2.5.1 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.5.1.1 Intensity interferometry for ultralight bosonic dark matter detection [55]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball

- **Posted on arXiv:** 05 February 2022

- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

2.5.1.2 One-Electron Quantum Cyclotron as a Milli-eV Dark-Photon Detector [56]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Roni Harnik, Thomas G. Myers, Harikrishnan Ramani, Benedict A.D. Sukra, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 12 August 2022
- **Published in Phys. Rev. Lett. 129, 261801 (2022):** 23 December 2022

2.5.1.3 Detection of bosenovae with quantum sensors on Earth and in space [57]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

2.5.1.4 Quantum entanglement of ions for light dark matter detection [58]

- **Authors:** Asuka Ito, Ryuichiro Kitano, Wakutaka Nakano, Ryoto Takai
- **Posted on arXiv:** 20 November 2023
- **Published in JHEP:** 16 February 2024

2.5.1.5 A Nuclear Interferometer for Ultra-Light Dark Matter Detection [59]

- **Authors:** Hannah Banks, Elina Fuchs, Matthew McCullough
- **Posted on arXiv:** 15 July 2024

2.5.1.6 Highly excited electron cyclotron for QCD axion and dark-photon detection [60]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Harikrishnan Ramani, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 07 October 2024
- **Published in Phys.Rev.D 111 (2025) 7, 075022:** 01 April 2025

2.5.1.7 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [61]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

2.5.1.8 Sensitively searching for microwave dark photons with atomic ensembles [62]

- **Authors:** Suirong He, De He, Yufen Li, Li Gao, Xianing Feng, Hao Zheng, L.F. Wei
- **Posted on arXiv:** 01 December 2024
- **Published in Phys.Rev.D:** 15 September 2025

2.5.1.9 Searches for exotic spin-dependent interactions with spin sensors [63]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

2.5.1.10 Spin squeezing of macroscopic nuclear spin ensembles [64]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

2.5.1.11 Detecting Dark Matter Using Optically Trapped Rydberg Atom Tweezer Arrays [65]

- **Authors:** So Chigusa, Taiyo Kasamaki, Toshi Kusano, Takeo Moroi, Kazunori Nakayama, Naoya Ozawa, Yoshiro Takahashi, Atsuhiro Umemoto, Amar Vutha
- **Posted on arXiv:** 17 July 2025
- **Published in Phys. Rev. Lett. 136, 151801 (2026):** 14 April 2026

2.5.1.12 Superradiant interactions for relic detection with entangled nuclear spins [66]

- **Authors:** Marios Galanis, Onur Hosten, Asimina Arvanitaki, Savas Dimopoulos
- **Posted on arXiv:** 28 August 2025
- **Published in Phys.Rev.A:** 10 June 2026

2.5.1.13 On the Speed-up of Wave-like Dark Matter Searches with Entangled Qubits [67]

- **Authors:** Arushi Bodas, Sohitri Ghosh, Roni Harnik
- **Posted on arXiv:** 13 October 2025

2.5.1.14 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [68]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

2.5.2 Quantum Sensors - Optical/Photonic Sensors

2.5.2.1 Laser Interferometers as Dark Matter Detectors [69]

- **Authors:** Evan D. Hall, Rana X. Adhikari, Valery V. Frolov, Holger Müller, Maxim Pospelov, Rana X Adhikari
- **Posted on arXiv:** 03 May 2016
- **Published in Phys.Rev.D:** 23 October 2018

2.5.2.2 Accelerating dark-matter axion searches with quantum measurement technology [70]

- **Authors:** Huaixiu Zheng, Matti Silveri, R.T. Brierley, S.M. Girvin, K.W. Lehnert
- **Posted on arXiv:** 08 July 2016

2.5.2.3 Searching for Dark Matter with a Superconducting Qubit [71]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

2.5.2.4 Detecting Hidden Photon Dark Matter Using the Direct Excitation of Transmon Qubits [72]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 07 December 2022
- **Published in Phys.Rev.Lett.:** 22 November 2023

2.5.2.5 Quantum measurements in fundamental physics: a user's manual [73]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023

2.5.2.6 Quantum Enhancement in Dark Matter Detection with Quantum Computation [74]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 17 November 2023
- **Published in Phys.Rev.Lett.:** 08 July 2024

2.5.2.7 Search for QCD axion dark matter with transmon qubits and quantum circuit [75]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 29 July 2024
- **Published in Phys.Rev.D:** 01 December 2024

2.5.2.8 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [76]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis
- **Posted on arXiv:** 07 August 2024
- **Published in Phys.Rev.D:** 01 March 2025

2.5.2.9 Axion bounds from quantum technology [77]

- **Authors:** Martin Bauer, Sreemanti Chakraborti, Guillaume Rostagni
- **Posted on arXiv:** 12 August 2024
- **Published in JHEP:** 05 May 2025

2.5.2.10 Quantum-enhanced dark matter detection with in-cavity control: mitigating the Rayleigh curse [78]

- **Authors:** Haowei Shi, Anthony J. Brady, Wojciech Górecki, Lorenzo Maccone, Roberto Di Candia, Quntao Zhuang
- **Posted on arXiv:** 06 September 2024
- **Published in npj Quantum Information 11, 48 (2025):** 17 March 2025

2.5.2.11 Transmon qubit modeling and characterization for Dark Matter search [79]

- **Authors:** Roberto Moretti, Danilo Labranca, Pietro Campana, Rodolfo Carobene, Marco Gobbo, Manuel A. Castellanos-Beltran, David Olaya, Peter F. Hopkins, Leonardo Banchi, Matteo Borghesi, Alessandro Candido, Stefano Carrazza, Hervè Atsè Corti, Alessandro D'Elia, Marco Faverzani, Elena Ferri, Angelo Nucciotti, Luca Origo, Andrea Pasquale, Alex Stephane Piedjou Komnang, Alessio Rettaroli, Simone Tocci, Claudio Gatti, Andrea Giachero
- **Posted on arXiv:** 09 September 2024
- **Published in IEEE Trans.Quantum Eng.:** 2026

2.5.2.12 Flux-Tunable Cavity for Dark Matter Detection [80]

- **Authors:** Fang Zhao, Ziqian Li, Akash V. Dixit, Tanay Roy, Andrei Vrajitoarea, Riju Banerjee, Alexander Anferov, Kan-Heng Lee, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 12 January 2025
- **Published in Phys.Rev.Lett.:** 13 November 2025

2.5.2.13 Quantum Enhanced Dark-Matter Search with Entangled Fock States in High-Quality Cavities [81]

- **Authors:** Benjamin Freiman, Xinyuan You, Andy C.Y. Li, Raphael Cervantes, Taeyoon Kim, Anna Grasselino, Roni Harnik, Yao Lu
- **Posted on arXiv:** 30 October 2025

2.5.2.14 Quantum Error Correction-like Noise Mitigation for Wave-like Dark Matter Searches with Quantum Sensors [82]

- **Authors:** Hajime Fukuda, Takeo Moroi, Thanaporn Sichanugrist
- **Posted on arXiv:** 05 November 2025

2.5.3 Quantum Sensors - Optomechanical Sensors

2.5.3.1 Sound of Dark Matter: Searching for Light Scalars with Resonant-Mass Detectors [83]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Ken Van Tilburg
- **Posted on arXiv:** 07 August 2015
- **Published in Phys.Rev.Lett.:** 22 January 2016

2.5.3.2 Dark Matter Direct Detection with Accelerometers [84]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

2.5.3.3 Ultralight dark matter detection with mechanical quantum sensors [85]

- **Authors:** Daniel Carney, Anson Hook, Zhen Liu, Jacob M. Taylor, Yue Zhao
- **Posted on arXiv:** 13 August 2019
- **Published in New Journal of Physics, Volume 23, February 2021:** 02 March 2021

2.5.3.4 Searching for scalar dark matter with compact mechanical resonators [86]

- **Authors:** Jack Manley, Russell Stump, Dalziel Wilson, Daniel Grin, Swati Singh
- **Posted on arXiv:** 16 October 2019
- **Published in Phys. Rev. Lett. 124, 151301 (2020):** 17 April 2020

2.5.3.5 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [87]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

2.5.3.6 Searching for vector dark matter with an optomechanical accelerometer [88]

- **Authors:** Jack Manley, Mitul Dey Chowdhury, Daniel Grin, Swati Singh, Dalziel J. Wilson
- **Posted on arXiv:** 09 July 2020
- **Published in Phys. Rev. Lett. 126, 061301 (2021):** 11 February 2021

2.5.3.7 Entanglement-enhanced optomechanical sensor array with application to dark matter searches [89]

- **Authors:** Anthony J. Brady, Xin Chen, Kewen Xiao, Yi Xia, Jack Manley, Mitul Dey Chowdhury, Zhen Liu, Roni Harnik, Dalziel J. Wilson, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 13 October 2022
- **Published in Commun.Phys.:** 01 September 2023

2.5.3.8 Axion detection with optomechanical cavities [90]

- **Authors:** Clara Murgui, Yikun Wang, Kathryn M. Zurek
- **Posted on arXiv:** 15 November 2022
- **Published in Phys.Lett.B:** 27 June 2024

2.5.3.9 Superfluid helium ultralight dark matter detector [91]

- **Authors:** M. Hirschel, V. Vadakkumbatt, N.P. Baker, F.M. Schweizer, J.C. Sankey, S. Singh, J.P. Davis
- **Posted on arXiv:** 14 September 2023
- **Published in Physical Review D 109, 095011 (2024):** 01 May 2024

2.5.3.10 Vector wave dark matter and terrestrial quantum sensors [92]

- **Authors:** Dorian W.P. Amaral, Mudit Jain, Mustafa A. Amin, Christopher Tunnell
- **Posted on arXiv:** 04 March 2024
- **Published in JCAP:** 21 June 2024

2.5.4 Quantum Sensors - Solid State Sensors

2.5.4.1 Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor [93]

- **Authors:** Xing Rong, Mengqi Wang, Jianpei Geng, Xi Qin, Maosen Guo, Man Jiao, Yijin Xie, Pengfei Wang, Pu Huang, Fazhan Shi, Yi-Fu Cai, Chongwen Zou, Jiangfeng Du
- **Posted on arXiv:** 12 June 2017
- **Published in Nature Commun.:** 21 February 2018

2.5.4.2 Axion search with quantum nondemolition detection of magnons [94]

- **Authors:** Tomonori Ikeda, Asuka Ito, Kentaro Miuchi, Jiro Soda, Hisaya Kurashige, Yutaka Shikano
- **Posted on arXiv:** 17 February 2021
- **Published in Phys. Rev. D 105, 102004 (2022):** 15 May 2022

2.5.4.3 Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors [95]

- **Authors:** P.-H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov, V.M. Acosta
- **Posted on arXiv:** 29 December 2021
- **Published in Physical Review Research 4, 023162 (2022):** 31 May 2022

2.5.4.4 Ultimate precision limit of noise sensing and dark matter search [96]

- **Authors:** Haowei Shi, Quntao Zhuang
- **Posted on arXiv:** 29 August 2022
- **Published in npj Quantum Inf.:** 20 March 2023

2.5.4.5 Light dark matter search with nitrogen-vacancy centers in diamonds [97]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 24 February 2023
- **Published in JHEP:** 12 March 2025

2.5.4.6 Axion detection via superfluid ^{3}He ferromagnetic phase and quantum measurement techniques [98]

- **Authors:** So Chigusa, Dan Kondo, Hitoshi Murayama, Risshin Okabe, Hiroyuki Sudo
- **Posted on arXiv:** 17 September 2023
- **Published in JHEP:** 26 September 2024

2.5.4.7 Nuclear spin metrology with nitrogen vacancy center in diamond for axion dark matter detection [99]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 09 July 2024
- **Published in Phys. Rev. D 111 (2025) 075028:** 01 April 2025

2.5.4.8 Listening for new physics with quantum acoustics [100]

- **Authors:** Ryan Linehan, Tanner Trickle, Christopher R. Conner, Sohitri Ghosh, Tongyan Lin, Mukul Sholapurkar, Andrew N. Cleland
- **Posted on arXiv:** 22 October 2024
- **Published in Phys.Rev.D:** 01 December 2025

2.5.4.9 Entanglement-enhanced ac magnetometry in the presence of Markovian noise [101]

- **Authors:** Thanaporn Sichanugrist, Hajime Fukuda, Takeo Moroi, Kazunori Nakayama, So Chigusa, Norikazu Mizuochi, Masashi Hazumi, Yuichiro Matsuzaki
- **Posted on arXiv:** 28 October 2024
- **Published in Phys.Rev.A:** 07 April 2025

2.5.4.10 Probing Sub-eV Dark Photon, Scalar and Axion-like Particle Dark Matters with Transmon Qubits [102]

- **Authors:** Wei Chao, Yu Gao, Ming-Jie Jin, Xiao-Sheng Liu, Xi-Lei Sun
- **Posted on arXiv:** 30 December 2024
- **Published in Phys.Dark Univ. 50 (2025) 102171:** 15 November 2025

2.5.4.11 Piezoelectric bulk acoustic resonators for dark photon detection [103]

- **Authors:** Tanner Trickle
- **Posted on arXiv:** 09 January 2025
- **Published in Phys.Rev.D:** 01 September 2025

2.5.4.12 Background suppression in quantum sensing of dark matter via collective entangled-state projection [104]

- **Authors:** Shion Chen, Hajime Fukuda, Yutaro Iiyama, Yuya Mino, Takeo Moroi, Mikio Nakahara, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 02 October 2025
- **Published in Phys.Rev.D:** 01 March 2026

2.5.4.13 Hybrid-spin decoupling for noise-resilient DC quantum sensing [105]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 20 November 2025

2.5.5 Crosslists

2.5.5.1 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

2.5.5.2 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

2.5.5.3 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [106]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

2.5.5.4 Mechanical quantum sensing in the search for dark matter [9]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.5.5.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

2.5.5.6 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

2.5.5.7 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie
- **Posted on arXiv:** 31 March 2022

2.5.5.8 Electromagnetic cavities as mechanical bars for gravitational waves [107]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D'Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

2.5.5.9 Quantum sensing for particle physics [108]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.5.5.10 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

2.5.5.11 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [109]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

2.6 Detector Technologies and Simulations

2.6.1 Continuous Variable Quantum Computing

2.6.1.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [110]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

2.6.2 Quantum Annealing

2.6.2.1 Unfolding measurement distributions via quantum annealing [111]

- **Authors:** Kyle Cormier, Riccardo Di Sipio, Peter Wittek
- **Posted on arXiv:** 22 August 2019
- **Published in JHEP:** 22 November 2019

2.6.3 Quantum Machine Learning - Unsupervised Methods

2.6.3.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [112]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

2.6.3.2 Running the Dual-PQC GAN on noisy simulators and real quantum hardware [113]

- **Authors:** Su Yeon Chang, Edwin Agnew, Elías F. Combarro, Michele Grossi, Steven Herbert, Sofia Vallecorsa
- **Posted on arXiv:** 30 May 2022
- **Published in J.Phys.Conf.Ser.:** 2023

2.6.3.3 Precise image generation on current noisy quantum computing devices [114]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borras, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

2.6.4 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

2.6.4.1 Quantum Systems for Enhanced High Energy Particle Physics Detectors [115]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

2.6.4.2 Quantum sensing for particle physics [108]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

2.6.5 Crosslists

2.6.5.1 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.6.5.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.6.5.3 Quantum-centric Supercomputing for Physics Research [116]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.7 Effective Field Theories

2.7.1 Quantum Algorithms - Quantum Simulations

2.7.1.1 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [117]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

2.7.2 Quantum Entanglement and Bell Inequalities

2.7.2.1 Minimal entanglement and emergent symmetries in low-energy QCD [118]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

2.7.3 Crosslists

2.7.3.1 Quantum SMEFT tomography: Top quark pair production at the LHC [119]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.7.3.2 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [11]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

2.7.3.3 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.8 Event Classification

2.8.1 Continuous Variable Quantum Computing

2.8.1.1 Continuous-variable photonic quantum extreme learning machines for fast collider-data selection [120]

- **Authors:** Benedikt Maier, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 October 2025
- **Published in EPJ Quant.Technol.:** 27 April 2026

2.8.2 Quantum Algorithms - Grover's Search Algorithm

2.8.2.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [121]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

2.8.3 Quantum Annealing

2.8.3.1 Solving a Higgs optimization problem with quantum annealing for machine learning [122]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

2.8.3.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [123]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

2.8.3.3 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [124]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

2.8.3.4 Quantum algorithm for the classification of supersymmetric top quark events [34]

- **Authors:** Pedrame Bargassa, Timothèe Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothèe Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.8.3.5 Completely quantum neural networks [125]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

2.8.4 Quantum Machine Learning - Supervised Methods

2.8.4.1 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [126]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevior
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

2.8.4.2 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [127]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

2.8.4.3 Higgs analysis with quantum classifiers [128]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

2.8.4.4 Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets [129]

- **Authors:** Zhelun Li, Lento Nagano, Koji Terashi
- **Posted on arXiv:** 17 May 2024
- **Published in Phys.Rev.Res.:** 10 October 2024

2.8.4.5 From Reachability to Learnability: Geometric Design Principles for Quantum Neural Networks [130]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky
- **Posted on arXiv:** 03 March 2026

2.8.5 Quantum Machine Learning - Unsupervised Methods

2.8.5.1 Guided graph compression for quantum graph neural networks [131]

- **Authors:** Mikel Casals, Vasilis Belis, Elias F. Combarro, Eduard Alarcón, Sofia Vallecorsa, Michele Grossi
- **Posted on arXiv:** 11 June 2025
- **Published in Mach.Learn.Sci.Tech.:** 12 September 2025

2.8.6 Quantum Machine Learning - Variational Quantum Algorithms

2.8.6.1 Event Classification with Quantum Machine Learning in High-Energy Physics [31]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.8.6.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [32]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.8.6.3 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [132]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

2.8.6.4 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [133]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

2.8.6.5 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [134]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

2.8.6.6 Quantum Metric Learning for New Physics Searches at the LHC [35]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

2.8.6.7 Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics [135]

- **Authors:** Eyup B. Unlu, Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva
- **Posted on arXiv:** 01 February 2024
- **Published in Axioms v. 13, no 3, (2024) 187:** 13 March 2024

2.8.6.8 Quantum Vision Transformers for Quark–Gluon Classification [136]

- **Authors:** Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva, Eyup B. Unlu
- **Posted on arXiv:** 16 May 2024
- **Published in Axioms 2024, 13(5), 323:** 13 May 2024

2.8.7 Crosslists

2.8.7.1 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.8.7.2 Unsupervised event classification with graphs on classical and photonic quantum computers [22]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.8.7.3 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.8.7.4 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.8.7.5 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.8.7.6 Quantum-centric Supercomputing for Physics Research [116]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.9 Event Generation

2.9.1 Quantum Algorithms - Grover's Search Algorithm

2.9.1.1 Quantum algorithm for Feynman loop integrals [137]

- **Authors:** Selomit Ramírez-Uribe, Andrès E. Rentería-Olivo, Germán Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

2.9.1.2 Loop Feynman integration on a quantum computer [138]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

2.9.2 Quantum Algorithms - Quantum Simulations

2.9.2.1 Quantum Algorithm for High Energy Physics Simulations [139]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

2.9.2.2 Towards a quantum computing algorithm for helicity amplitudes and parton showers [140]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

2.9.2.3 Improving quantum simulation efficiency of final state radiation with dynamic quantum circuits [141]

- **Authors:** Plato Deliyannis, James Sud, Diana Chamaki, Zoë Webb-Mack, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 18 March 2022
- **Published in Physical Review D 106, 036007 (2022):** 01 August 2022

2.9.2.4 Collider events on a quantum computer [142]

- **Authors:** Gösta Gustafson, Stefan Prestel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 21 July 2022
- **Published in JHEP:** 07 November 2022

2.9.2.5 Quantum parton shower with kinematics [143]

- **Authors:** Christian W. Bauer, So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 30 October 2023
- **Published in Phys.Rev.A:** 26 March 2024

2.9.3 Quantum Algorithms - Quantum Walks

2.9.3.1 Quantum walk approach to simulating parton showers [144]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

2.9.4 Quantum Machine Learning - Unsupervised Methods

2.9.4.1 Style-based quantum generative adversarial networks for Monte Carlo events [145]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

2.9.4.2 Quantum integration of elementary particle processes [146]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

2.9.4.3 Generative invertible quantum neural networks [147]

- **Authors:** Armand Rousselot, Michael Spannowsky
- **Posted on arXiv:** 24 February 2023
- **Published in SciPost Phys. 16, 146 (2024):** 04 June 2024

2.9.4.4 Quantum-centric Supercomputing for Physics Research [116]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

2.9.5 Quantum Machine Learning - Variational Quantum Algorithms

2.9.5.1 Partonic collinear structure by quantum computing [148]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.9.5.2 Unsupervised quantum circuit learning in high energy physics [149]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.9.5.3 Quantum Wishlist: Lessons from Parton Showers [150]

- **Authors:** Masahito Yamazaki
- **Posted on arXiv:** 25 February 2025
- **Published in PoS:** 24 October 2025

2.9.6 Crosslists

2.9.6.1 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.9.6.2 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti

- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.9.6.3 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.9.6.4 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

2.9.6.5 Quantum data learning for quantum simulations in high-energy physics [151]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.9.6.6 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.9.6.7 Quantum Algorithms in Particle Physics [152]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.10 Gravitational Waves

2.10.1 Quantum Sensors - Optical/Photonic Sensors

2.10.1.1 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [106]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

2.10.1.2 Electromagnetic cavities as mechanical bars for gravitational waves [107]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D'Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

2.10.1.3 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [109]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

2.10.2 Quantum Sensors - Optomechanical Sensors

2.10.2.1 Quantum interactions between a laser interferometer and gravitational waves [153]

- **Authors:** Belinda Pang, Yanbei Chen
- **Posted on arXiv:** 28 August 2018
- **Published in Phys. Rev. D 98, 124006 (2018):** 11 December 2018

2.10.2.2 Superconducting Levitated Detector of Gravitational Waves [154]

- **Authors:** Daniel Carney, Gerard Higgins, Giacomo Marocco, Michael Wentzel
- **Posted on arXiv:** 02 August 2024
- **Published in Carney (2025) Phys. Rev. Lett. 134, 181402:** 06 May 2025

2.11 Jet Reconstruction

2.11.1 Quantum Algorithms - Grover's Search Algorithm

2.11.1.1 Quantum Algorithms for Jet Clustering [155]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

2.11.1.2 Quantum Algorithms in Particle Physics [152]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

2.11.2 Quantum Annealing

2.11.2.1 Adiabatic quantum algorithm for multijet clustering in high energy physics [156]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

2.11.2.2 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [157]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

2.11.2.3 Quantum annealing for jet clustering with thrust [158]

- **Authors:** Andrea Delgado, Jesse Thaler
- **Posted on arXiv:** 05 May 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.11.2.4 Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics [159]

- **Authors:** Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, Jesse Thaler
- **Posted on arXiv:** 20 May 2022
- **Published in Phys. Rev. D 106, 056008 (2022):** 01 September 2022

2.11.3 Quantum Machine Learning - Supervised Methods

2.11.3.1 Quantum-inspired machine learning on high-energy physics data [160]

- **Authors:** Timo Felser, Marco Trenti, Lorenzo Sestini, Alessio Gianelle, Davide Zuliani, Donatella Lucchesi, Simone Montangero
- **Posted on arXiv:** 28 April 2020
- **Published in npj Quantum Inf.:** 15 July 2021

2.11.3.2 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [161]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

2.11.3.3 Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data [162]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 21 February 2022
- **Published in Phys. Rev. A 106 (Dec, 2022) 062423:** 19 December 2022

2.11.4 Quantum Machine Learning - Unsupervised Methods

2.11.4.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [163]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

2.11.4.2 Quantum clustering and jet reconstruction at the LHC [164]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

2.11.5 Quantum Machine Learning - Variational Quantum Algorithms

2.11.5.1 Quantum Machine Learning for b-jet charge identification [165]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

2.11.6 Crosslists

2.11.6.1 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.11.6.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.12 Lattice Field Theories - Scalar/Fermion Theories

2.12.1 Continuous Variable Quantum Computing

2.12.1.1 Simulating quantum field theories on continuous-variable quantum computers [166]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 March 2024
- **Published in Phys. Rev. A 110 (2024) 1, 012607:** 08 July 2024

2.12.1.2 Real-time scattering processes with continuous-variable quantum computers [167]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 03 February 2025
- **Published in Phys. Rev. A 112 (2025), 012614:** 16 July 2025

2.12.1.3 Qumode Tensor Networks for False Vacuum Decay in Quantum Field Theory [168]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 20 June 2025

2.12.2 Quantum Algorithms - Quantum Simulations

2.12.2.1 Quantum Computation of Scattering in Scalar Quantum Field Theories [169]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

2.12.2.2 Quantum Algorithms for Fermionic Quantum Field Theories [170]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

2.12.2.3 Digitization of scalar fields for quantum computing [171]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

2.12.2.4 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [172]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

2.12.2.5 Real-Time Scattering on Quantum Computers via Hamiltonian Truncation [173]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams, Matthew Wingate
- **Posted on arXiv:** 06 May 2025

2.13 Lattice Field Theories - Gauge Theories

2.13.1 Quantum Algorithms - Quantum Simulations

2.13.1.1 Simulating lattice gauge theories on a quantum computer [174]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

2.13.1.2 Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories [175]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 07 May 2013
- **Published in Annalen Phys.:** 2013

2.13.1.3 Towards Quantum Simulating QCD [176]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

2.13.1.4 Quantum Simulations of Lattice Gauge Theories using Ultracold Atoms in Optical Lattices [177]

- **Authors:** Erez Zohar, J. Ignacio Cirac, Benni Reznik
- **Posted on arXiv:** 08 March 2015
- **Published in Rept.Prog.Phys.:** 18 December 2015

2.13.1.5 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [178]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

2.13.1.6 General Methods for Digital Quantum Simulation of Gauge Theories [179]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

2.13.1.7 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [180]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

2.13.1.8 Simulating Lattice Gauge Theories within Quantum Technologies [181]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

2.13.1.9 Toward scalable simulations of Lattice Gauge Theories on quantum computers [182]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

2.13.1.10 Role of boundary conditions in quantum computations of scattering observables [183]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

2.13.1.11 Quantum simulation of gauge theory via orbifold lattice [184]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

2.13.1.12 Lattice renormalization of quantum simulations [185]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

2.13.1.13 Lattice Quantum Chromodynamics and Electrodynamics on a Universal Quantum Computer [186]

- **Authors:** Angus Kan, Yunseong Nam
- **Posted on arXiv:** 27 July 2021

2.13.1.14 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [187]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

2.13.1.15 Quantum error correction with gauge symmetries [188]

- **Authors:** Abhishek Rajput, Alessandro Roggero, Nathan Wiebe
- **Posted on arXiv:** 09 December 2021
- **Published in npj Quantum Inf.:** 25 April 2023

2.13.1.16 Preparations for quantum simulations of quantum chromodynamics in <math display=”inline”><mrow><mn>1</mn><mo>+</mo><mn>1</mn></mrow></ dimensions. I. Axial gauge [189]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

2.13.1.17 Measurement-based quantum simulation of Abelian lattice gauge theories [190]

- **Authors:** Hiroki Sukeno, Takuya Okuda
- **Posted on arXiv:** 19 October 2022
- **Published in SciPost Phys. 14, 129 (2023):** 25 May 2023

2.13.1.18 General quantum algorithms for Hamiltonian simulation with applications to a non-Abelian lattice gauge theory [191]

- **Authors:** Zohreh Davoudi, Alexander F. Shaw, Jesse R. Stryker
- **Posted on arXiv:** 28 December 2022
- **Published in Quantum 7, 1213 (2023):** 20 December 2023

2.13.1.19 Simulating <math><msub><mi mathvariant=”double-struck”>Z</mi><mn>2</m lattice gauge theory on a quantum computer [192]

- **Authors:** Clement Charles, Erik J. Gustafson, Elizabeth Hardt, Florian Herren, Norman Hogan, Henry Lamm, Sara Starecheski, Ruth S. Van de Water, Michael L. Wagman
- **Posted on arXiv:** 03 May 2023
- **Published in Phys.Rev.E:** 26 January 2024

2.13.1.20 Quantum simulation of fundamental particles and forces [193]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 April 2024
- **Published in Nature Rev.Phys.:** 21 June 2023

2.13.1.21 Toward QCD on quantum computer: orbifold lattice approach [194]

- **Authors:** Georg Bergner, Masanori Hanada, Enrico Rinaldi, Andreas Schafer
- **Posted on arXiv:** 22 January 2024
- **Published in JHEP:** 20 May 2024

2.13.1.22 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [195]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

2.13.1.23 Quantum Simulation of SU(3) Lattice Yang-Mills Theory at Leading Order in Large-<math display=”inline”><msub><mi>N</mi><mi>c</mi></msub></mat Expansion [196]

- **Authors:** Anthony N. Ciavarella, Christian W. Bauer
- **Posted on arXiv:** 15 February 2024
- **Published in Phys.Rev.Lett.:** 09 September 2024

2.13.1.24 Scalable cold-atom quantum simulator of a $3+1$D U(1) lattice gauge theory with dynamical matter [197]

- **Authors:** Simone Orlando, Guo-Xian Su, Bing Yang, Jad C. Halimeh
- **Posted on arXiv:** 07 January 2026

2.13.2 Quantum Annealing

2.13.2.1 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [198]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

2.13.2.2 SU(2) lattice gauge theory on a quantum annealer [199]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

2.13.3 Quantum Machine Learning - Supervised Methods

2.13.3.1 Quantum data learning for quantum simulations in high-energy physics [151]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.13.4 Quantum Machine Learning - Variational Quantum Algorithms

2.13.4.1 SU(2) hadrons on a quantum computer via a variational approach [200]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

2.13.5 Crosslists

2.13.5.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [201]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

2.13.5.2 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

2.13.5.3 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [10]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

2.13.5.4 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

2.13.5.5 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

2.13.5.6 Quantum computing hardware for HEP algorithms and sensing [18]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti
- **Posted on arXiv:** 18 April 2022

2.13.5.7 Review on Quantum Computing for Lattice Field Theory [5]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in PoS:** 01 February 2023

2.13.5.8 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico

Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

2.13.5.9 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [202]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

2.13.5.10 Steps toward quantum simulations of hadronization and energy loss in dense matter [203]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

2.14 Lattice Field Theories - Schwinger Model

2.14.1 Quantum Algorithms - Quantum Simulations

2.14.1.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [201]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

2.14.1.2 Quantum Algorithms for Simulating the Lattice Schwinger Model [204]

- **Authors:** Alexander F. Shaw, Pavel Lougovski, Jesse R. Stryker, Nathan Wiebe
- **Posted on arXiv:** 25 February 2020
- **Published in Quantum 4, 306 (2020):** 10 August 2020

2.14.1.3 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [202]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

2.14.1.4 Steps toward quantum simulations of hadronization and energy loss in dense matter [203]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

2.14.1.5 Enhancing quantum field theory simulations on NISQ devices with Hamiltonian truncation [205]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams
- **Posted on arXiv:** 26 July 2024
- **Published in Phys. Rev. D 110 (2024), 096016:** 01 November 2024

2.14.2 Quantum Machine Learning - Variational Quantum Algorithms

2.14.2.1 Quantum-classical computation of Schwinger model dynamics using quantum computers [206]

- **Authors:** N. Klco, E.F. Dumitrescu, A.J. McCaskey, T.D. Morris, R.C. Pooser, M. Sanz, E. Solano, P. Lougovski, M.J. Savage
- **Posted on arXiv:** 08 March 2018
- **Published in Phys. Rev. A 98, 032331 (2018):** 29 September 2018

2.14.2.2 Quantum-Classical Simulation of Quantum Field Theory by Quantum Circuit Learning [207]

- **Authors:** Kazuki Ikeda
- **Posted on arXiv:** 27 November 2023
- **Published in Annalen Phys.:** 01 June 2025

2.14.3 Crosslists

2.14.3.1 Quantum data learning for quantum simulations in high-energy physics [151]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

2.15 Quantum Chromodynamics

2.15.1 Quantum Algorithms - Quantum Simulations

2.15.1.1 Quantum simulation of quantum field theory in the light-front formulation [208]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

2.15.1.2 Quantum simulation of colour in perturbative quantum chromodynamics [209]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen
- **Posted on arXiv:** 08 March 2023
- **Published in SciPost Phys. 15, 205 (2023):** 24 November 2023

2.15.1.3 Quantum simulation of scattering amplitudes and interferences in perturbative QCD [210]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen, Simon Williams
- **Posted on arXiv:** 09 July 2025

2.15.2 Quantum Entanglement and Bell Inequalities

2.15.2.1 Symmetry from entanglement suppression [211]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

2.15.3 Crosslists

2.15.3.1 Partonic collinear structure by quantum computing [148]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.15.3.2 Minimal entanglement and emergent symmetries in low-energy QCD [118]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

2.16 Quantum Correlations at Colliders

2.16.1 Quantum Entanglement and Bell Inequalities

2.16.1.1 Entanglement and quantum tomography with top quarks at the LHC [212]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

2.16.1.2 Testing Bell Inequalities at the LHC with Top-Quark Pairs [213]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

2.16.1.3 Quantum tops at the LHC: from entanglement to Bell inequalities [214]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

2.16.1.4 Quantum SMEFT tomography: Top quark pair production at the LHC [119]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.16.1.5 Quantum information with top quarks in QCD [215]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

2.16.1.6 Improved tests of entanglement and Bell inequalities with LHC tops [216]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
- **Posted on arXiv:** 01 May 2022
- **Published in Eur.Phys.J.C:** 03 August 2022

2.16.1.7 Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs [217]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli
- **Posted on arXiv:** 24 August 2022
- **Published in Eur.Phys.J.C:** 19 February 2023

2.16.1.8 Quantum Discord and Steering in Top Quarks at the LHC [218]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 08 September 2022
- **Published in Phys. Rev. Lett. 130, 221801 (2023):** 31 May 2023

2.16.1.9 Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders [219]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 01 February 2023
- **Published in Eur.Phys.J.C:** 15 September 2023

2.16.1.10 Three-Body Entanglement in Particle Decays [220]

- **Authors:** Kazuki Sakurai, Michael Spannowsky
- **Posted on arXiv:** 02 October 2023
- **Published in Phys.Rev.Lett.:** 09 April 2024

2.16.1.11 Can Bell inequalities be tested via scattering cross-section at colliders ? [221]

- **Authors:** Song Li, Wei Shen, Jin Min Yang
- **Posted on arXiv:** 02 January 2024
- **Published in Eur.Phys.J.C:** 21 November 2024

2.16.1.12 Quantum detection of new physics in top-quark pair production at the LHC [222]

- **Authors:** Fabio Maltoni, Claudio Severi, Simone Tentori, Eleni Vryonidou
- **Posted on arXiv:** 16 January 2024
- **Published in JHEP:** 15 March 2024

2.16.1.13 Full quantum tomography of top quark decays [223]

- **Authors:** J.A. Aguilar-Saavedra
- **Posted on arXiv:** 22 February 2024
- **Published in Phys.Lett.B:** 04 July 2024

2.16.1.14 Quantum information meets high-energy physics: input to the update of the European strategy for particle physics [224]

- **Authors:** Yoav Afik, Federica Fabbri, Matthew Low, Luca Marzola, Juan Antonio Aguilar-Saavedra, Mohammad Mahdi Altakach, Nedaa Alexandra Asbah, Yang Bai, Hannah Banks, Alan J. Barr, Alexander Bernal, Thomas E. Browder, Paweł Caban, J. Alberto Casas, Kun Cheng, Frèdèric Dèliot, Regina Demina, Antonio Di Domenico, Michał Eckstein, Marco Fabbrichesi, Benjamin Fuks, Emidio Gabrielli, Dorival Gonçalves, Radosław Grabarczyk, Michele Grossi, Tao Han, Timothy J. Hobbs, Paweł Horodecki, James Howarth, Shih-Chieh Hsu, Stephen Jiggins, Eleanor Jones, Andreas W. Jung, Andrea Helen Knue, Steffen Korn, Theodota Lagouri, Priyanka Lamba, Gabriel T. Landi, Haifeng Li, Qiang Li, Ian Low, Fabio Maltoni, Josh McFayden, Navin McGinnis, Roberto A. Morales, Jesús M. Moreno, Juan Ramón Muñoz de Nova, Giulia Negro, Davide Pagani, Giovanni Pelliccioli, Michele Pinamonti, Laura Pintucci, Baptiste Ravina, Alim Ruzi, Kazuki Sakurai, Ethan Simpson, Maximiliano Sioli, Shufang Su, Sokratis Trifinopoulos, Sven E. Vahsen, Sofia Vallecorsa, Alessandro Vicini, Marcel Vos, Eleni Vryonidou, Chris D. White, Martin J. White, Andrew J. Wildridge, Tong Arthur Wu, Laura Zani, Yulei Zhang, Knut Zoch
- **Posted on arXiv:** 31 March 2025
- **Published in Eur.Phys.J.Plus:** 09 September 2025

2.16.1.15 Colliders are Testing neither Locality via Bell's Inequality nor Entanglement versus Non-Entanglement [225]

- **Authors:** Steven A. Abel, Herbi K. Dreiner, Rhitaja Sengupta, Lorenzo Ubaldi
- **Posted on arXiv:** 21 July 2025

2.16.2 Crosslists

2.16.2.1 Quantum entanglement and Bell inequality violation at colliders [7]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

2.17 Track Reconstruction

2.17.1 Quantum Algorithms - Grover's Search Algorithm

2.17.1.1 Quantum Associative Memory in HEP Track Pattern Recognition [226]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

2.17.1.2 Quantum speedup for track reconstruction in particle accelerators [227]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

2.17.1.3 Quantum pathways for charged track finding in high-energy collisions [228]

- **Authors:** Christopher Brown, Michael Spannowsky, Alexander Tapper, Simon Williams, Ioannis Xiotidis
- **Posted on arXiv:** 01 November 2023
- **Published in Front. Artif. Intell. 7:1339785 (2024):** 07 May 2024

2.17.2 Quantum Algorithms - Harrow-Hassidim-Lloyd Algorithm

2.17.2.1 A quantum algorithm for track reconstruction in the LHCb vertex detector [229]

- **Authors:** Davide Nicotra, Miriam Lucio Martinez, Jacco Andreas de Vries, Marcel Merk, Kurt Driessens, Ronald Leonard Westra, Domenica Dibenedetto, Daniel Hugo Cámpora Pèrez
- **Posted on arXiv:** 01 August 2023
- **Published in JINST:** 29 November 2023

2.17.3 Quantum Annealing

2.17.3.1 A pattern recognition algorithm for quantum annealers [230]

- **Authors:** Frèdèric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
- **Published in Comput.Softw.Big Sci.:** 09 December 2019

2.17.3.2 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [231]

- **Authors:** Souvik Das, Andrew J. Wildridge, Andreas Jung
- **Posted on arXiv:** 21 March 2019

2.17.3.3 Charged particle tracking with quantum annealing optimization [232]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

2.17.3.4 Quantum annealing algorithms for track pattern recognition [233]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

2.17.3.5 Particle track classification using quantum associative memory [234]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

2.17.3.6 Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders [235]

- **Authors:** Hideki Okawa, Qing-Guo Zeng, Xian-Zhe Tao, Man-Hong Yung
- **Posted on arXiv:** 22 February 2024
- **Published in Comput.Softw.Big Sci.:** 28 August 2024

2.17.4 Quantum Machine Learning - Supervised Methods

2.17.4.1 Reconstructing charged particle track segments with a quantum-enhanced support vector machine [236]

- **Authors:** Philippa Duckett, Gabriel Facini, Marcin Jastrzebski, Sarah Malik, Tim Scanlon, Sebastien Rettie
- **Posted on arXiv:** 14 December 2022
- **Published in Phys.Rev.D:** 01 March 2024

2.17.5 Quantum Machine Learning - Variational Quantum Algorithms

2.17.5.1 Quantum convolutional neural networks for high energy physics data analysis [33]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.17.5.2 Hybrid quantum classical graph neural networks for particle track reconstruction [237]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

2.17.5.3 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [238]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in J.Phys.Conf.Ser.:** 2023

2.17.5.4 Particle track reconstruction with noisy intermediate-scale quantum computers [239]

- **Authors:** Tim Schwägerl, Cigdem Issever, Karl Jansen, Teng Jian Khoo, Stefan Kühn, Cenk Tüysüz, Hannsjörg Weber
- **Posted on arXiv:** 23 March 2023
- **Published in J.Phys.Conf.Ser.:** 2026

2.17.5.5 Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm [240]

- **Authors:** Hideki Okawa
- **Posted on arXiv:** 16 October 2023

2.17.6 Crosslists

2.17.6.1 Hybrid Quantum-Classical Graph Convolutional Network [30]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

2.17.6.2 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

2.17.6.3 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

2.17.6.4 Quantum Systems for Enhanced High Energy Particle Physics Detectors [115]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

2.17.6.5 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang
- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3 Quantum Information Science in Nuclear and Particle Physics

3.1 Reviews and Whitepapers and Proceedings

3.1.1 Reviews

3.1.1.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

3.1.1.2 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

3.1.1.3 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.1.1.4 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.1.1.5 Review on Quantum Computing for Lattice Field Theory [5]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in PoS:** 01 February 2023

3.1.1.6 Machine learning for anomaly detection in particle physics [6]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klaeboe Aarrestad
- **Posted on arXiv:** 20 December 2023
- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

3.1.1.7 Quantum entanglement and Bell inequality violation at colliders [7]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

3.1.2 Whitepapers and Proceedings

3.1.2.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

3.1.2.2 Mechanical quantum sensing in the search for dark matter [9]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

3.1.2.3 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [10]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

3.1.2.4 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [11]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.1.2.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.1.2.6 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.1.2.7 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.1.2.8 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.1.2.9 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.1.2.10 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti

- **Posted on arXiv:** 07 April 2022

- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

3.1.2.11 Quantum computing hardware for HEP algorithms and sensing [18]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti

- **Posted on arXiv:** 18 April 2022

3.1.2.12 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage

- **Posted on arXiv:** 14 September 2022

3.1.2.13 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023

- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.1.2.14 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik,

J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

3.2 Continuous Variable Quantum Computing

3.2.1 Anomaly Detection

3.2.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [22]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

3.2.2 Detector Technologies and Simulations

3.2.2.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [110]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.2.3 Event Classification

3.2.3.1 Continuous-variable photonic quantum extreme learning machines for fast collider-data selection [120]

- **Authors:** Benedikt Maier, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 October 2025
- **Published in EPJ Quant.Technol.:** 27 April 2026

3.2.4 Lattice Field Theories - Scalar/Fermion Theories

3.2.4.1 Simulating quantum field theories on continuous-variable quantum computers [166]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 March 2024
- **Published in Phys. Rev. A 110 (2024) 1, 012607:** 08 July 2024

3.2.4.2 Real-time scattering processes with continuous-variable quantum computers [167]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 03 February 2025
- **Published in Phys. Rev. A 112 (2025), 012614:** 16 July 2025

3.2.4.3 Qumode Tensor Networks for False Vacuum Decay in Quantum Field Theory [168]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 20 June 2025

3.3 Quantum Algorithms - Grover's Search Algorithm

3.3.1 Beyond the Standard Model

3.3.1.1 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [28]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

3.3.2 Event Classification

3.3.2.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [121]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

3.3.3 Event Generation

3.3.3.1 Quantum algorithm for Feynman loop integrals [137]

- **Authors:** Selomit Ramírez-Uribe, Andrès E. Rentería-Olivo, Germán Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

3.3.3.2 Loop Feynman integration on a quantum computer [138]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Michele Grossi, Sofia Vallecorsa, Germán Rodrigo
- **Posted on arXiv:** 05 January 2024
- **Published in Phys. Rev. D 110, 074031 (2024):** 01 October 2024

3.3.4 Jet Reconstruction

3.3.4.1 Quantum Algorithms for Jet Clustering [155]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.3.4.2 Quantum Algorithms in Particle Physics [152]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.3.5 Track Reconstruction

3.3.5.1 Quantum Associative Memory in HEP Track Pattern Recognition [226]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

3.3.5.2 Quantum speedup for track reconstruction in particle accelerators [227]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

3.3.5.3 Quantum pathways for charged track finding in high-energy collisions [228]

- **Authors:** Christopher Brown, Michael Spannowsky, Alexander Tapper, Simon Williams, Ioannis Xiotidis
- **Posted on arXiv:** 01 November 2023
- **Published in Front. Artif. Intell. 7:1339785 (2024):** 07 May 2024

3.3.6 Crosslists

3.3.6.1 Quantum clustering and jet reconstruction at the LHC [164]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

3.4 Quantum Algorithms - Harrow-Hassidim-Lloyd Algorithm

3.4.1 Track Reconstruction

3.4.1.1 A quantum algorithm for track reconstruction in the LHCb vertex detector [229]

- **Authors:** Davide Nicotra, Miriam Lucio Martinez, Jacco Andreas de Vries, Marcel Merk, Kurt Driessens, Ronald Leonard Westra, Domenica Dibenedetto, Daniel Hugo Cámpora Pèrez
- **Posted on arXiv:** 01 August 2023
- **Published in JINST:** 29 November 2023

3.4.2 Crosslists

3.4.2.1 Quantum-centric Supercomputing for Physics Research [116]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.5 Quantum Algorithms - Quantum Simulations

3.5.1 Dark Matter - Particle-like Dark Matter

3.5.1.1 Quantum simulations of dark sector showers [36]

- **Authors:** So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 26 April 2022
- **Published in Phys.Lett.B:** 26 September 2022

3.5.2 Effective Field Theories

3.5.2.1 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [117]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

3.5.3 Event Generation

3.5.3.1 Quantum Algorithm for High Energy Physics Simulations [139]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

3.5.3.2 Towards a quantum computing algorithm for helicity amplitudes and parton showers [140]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

3.5.3.3 Improving quantum simulation efficiency of final state radiation with dynamic quantum circuits [141]

- **Authors:** Plato Deliyannis, James Sud, Diana Chamaki, Zoë Webb-Mack, Christian W. Bauer, Benjamin Nachman
- **Posted on arXiv:** 18 March 2022
- **Published in Physical Review D 106, 036007 (2022):** 01 August 2022

3.5.3.4 Collider events on a quantum computer [142]

- **Authors:** Gösta Gustafson, Stefan Prestel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 21 July 2022
- **Published in JHEP:** 07 November 2022

3.5.3.5 Quantum parton shower with kinematics [143]

- **Authors:** Christian W. Bauer, So Chigusa, Masahito Yamazaki
- **Posted on arXiv:** 30 October 2023
- **Published in Phys.Rev.A:** 26 March 2024

3.5.4 Lattice Field Theories - Scalar/Fermion Theories

3.5.4.1 Quantum Computation of Scattering in Scalar Quantum Field Theories [169]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

3.5.4.2 Quantum Algorithms for Fermionic Quantum Field Theories [170]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

3.5.4.3 Digitization of scalar fields for quantum computing [171]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

3.5.4.4 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [172]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

3.5.4.5 Real-Time Scattering on Quantum Computers via Hamiltonian Truncation [173]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams, Matthew Wingate
- **Posted on arXiv:** 06 May 2025

3.5.5 Lattice Field Theories - Gauge Theories

3.5.5.1 Simulating lattice gauge theories on a quantum computer [174]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

3.5.5.2 Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories [175]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 07 May 2013
- **Published in Annalen Phys.:** 2013

3.5.5.3 Towards Quantum Simulating QCD [176]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

3.5.5.4 Quantum Simulations of Lattice Gauge Theories using Ultracold Atoms in Optical Lattices [177]

- **Authors:** Erez Zohar, J. Ignacio Cirac, Benni Reznik
- **Posted on arXiv:** 08 March 2015
- **Published in Rept.Prog.Phys.:** 18 December 2015

3.5.5.5 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [178]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

3.5.5.6 General Methods for Digital Quantum Simulation of Gauge Theories [179]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

3.5.5.7 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [180]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

3.5.5.8 Simulating Lattice Gauge Theories within Quantum Technologies [181]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.5.5.9 Toward scalable simulations of Lattice Gauge Theories on quantum computers [182]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

3.5.5.10 Role of boundary conditions in quantum computations of scattering observables [183]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

3.5.5.11 Quantum simulation of gauge theory via orbifold lattice [184]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

3.5.5.12 Lattice renormalization of quantum simulations [185]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

3.5.5.13 Lattice Quantum Chromodynamics and Electrodynamics on a Universal Quantum Computer [186]

- **Authors:** Angus Kan, Yunseong Nam
- **Posted on arXiv:** 27 July 2021

3.5.5.14 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [187]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

3.5.5.15 Quantum error correction with gauge symmetries [188]

- **Authors:** Abhishek Rajput, Alessandro Roggero, Nathan Wiebe
- **Posted on arXiv:** 09 December 2021
- **Published in npj Quantum Inf.:** 25 April 2023

3.5.5.16 Preparations for quantum simulations of quantum chromodynamics in <math display=”inline”><mrow><mn>1</mn><mo>+</mo><mn>1</mn></mrow></ dimensions. I. Axial gauge [189]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

3.5.5.17 Measurement-based quantum simulation of Abelian lattice gauge theories [190]

- **Authors:** Hiroki Sukeno, Takuya Okuda
- **Posted on arXiv:** 19 October 2022
- **Published in SciPost Phys. 14, 129 (2023):** 25 May 2023

3.5.5.18 General quantum algorithms for Hamiltonian simulation with applications to a non-Abelian lattice gauge theory [191]

- **Authors:** Zohreh Davoudi, Alexander F. Shaw, Jesse R. Stryker
- **Posted on arXiv:** 28 December 2022
- **Published in Quantum 7, 1213 (2023):** 20 December 2023

3.5.5.19 Simulating <math><msub><mi mathvariant=”double-struck”>Z</mi><mn>2</m lattice gauge theory on a quantum computer [192]

- **Authors:** Clement Charles, Erik J. Gustafson, Elizabeth Hardt, Florian Herren, Norman Hogan, Henry Lamm, Sara Starecheski, Ruth S. Van de Water, Michael L. Wagman
- **Posted on arXiv:** 03 May 2023
- **Published in Phys.Rev.E:** 26 January 2024

3.5.5.20 Quantum simulation of fundamental particles and forces [193]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 09 April 2024
- **Published in Nature Rev.Phys.:** 21 June 2023

3.5.5.21 Toward QCD on quantum computer: orbifold lattice approach [194]

- **Authors:** Georg Bergner, Masanori Hanada, Enrico Rinaldi, Andreas Schafer
- **Posted on arXiv:** 22 January 2024
- **Published in JHEP:** 20 May 2024

3.5.5.22 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [195]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

3.5.5.23 Quantum Simulation of SU(3) Lattice Yang-Mills Theory at Leading Order in Large-<math display=”inline”><msub><mi>N</mi><mi>c</mi></msub></math> Expansion [196]

- **Authors:** Anthony N. Ciavarella, Christian W. Bauer
- **Posted on arXiv:** 15 February 2024
- **Published in Phys.Rev.Lett.:** 09 September 2024

3.5.5.24 Scalable cold-atom quantum simulator of a $3 + 1$D U(1) lattice gauge theory with dynamical matter [197]

- **Authors:** Simone Orlando, Guo-Xian Su, Bing Yang, Jad C. Halimeh
- **Posted on arXiv:** 07 January 2026

3.5.6 Lattice Field Theories - Schwinger Model

3.5.6.1 Quantum simulation of a lattice Schwinger model in a chain of trapped ions [201]

- **Authors:** Philipp Hauke, David Marcos, Marcello Dalmonte, Peter Zoller
- **Posted on arXiv:** 10 June 2013
- **Published in Phys.Rev.X:** 22 November 2013

3.5.6.2 Quantum Algorithms for Simulating the Lattice Schwinger Model [204]

- **Authors:** Alexander F. Shaw, Pavel Lougovski, Jesse R. Stryker, Nathan Wiebe
- **Posted on arXiv:** 25 February 2020
- **Published in Quantum 4, 306 (2020):** 10 August 2020

3.5.6.3 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [202]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

3.5.6.4 Steps toward quantum simulations of hadronization and energy loss in dense matter [203]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

3.5.6.5 Enhancing quantum field theory simulations on NISQ devices with Hamiltonian truncation [205]

- **Authors:** James Ingoldby, Michael Spannowsky, Timur Sypchenko, Simon Williams
- **Posted on arXiv:** 26 July 2024
- **Published in Phys. Rev. D 110 (2024), 096016:** 01 November 2024

3.5.7 Quantum Chromodynamics

3.5.7.1 Quantum simulation of quantum field theory in the light-front formulation [208]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

3.5.7.2 Quantum simulation of colour in perturbative quantum chromodynamics [209]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen
- **Posted on arXiv:** 08 March 2023
- **Published in SciPost Phys. 15, 205 (2023):** 24 November 2023

3.5.7.3 Quantum simulation of scattering amplitudes and interferences in perturbative QCD [210]

- **Authors:** Herschel A. Chawdhry, Mathieu Pellen, Simon Williams
- **Posted on arXiv:** 09 July 2025

3.5.8 Crosslists

3.5.8.1 Lattice gauge theory simulations in the quantum information era [1]

- **Authors:** M. Dalmonte, S. Montangero
- **Posted on arXiv:** 11 February 2016
- **Published in Contemporary Physics 57 388 (2016):** 09 March 2016

3.5.8.2 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [10]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

3.5.8.3 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [11]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.5.8.4 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker
- **Posted on arXiv:** 14 March 2022

3.5.8.5 Quantum Simulation for High-Energy Physics [17]

- **Authors:** Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J. Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, Silvia Zorzetti
- **Posted on arXiv:** 07 April 2022
- **Published in PRX Quantum 4, 027001, 2023:** 01 May 2023

3.5.8.6 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.5.8.7 Review on Quantum Computing for Lattice Field Theory [5]

- **Authors:** Lena Funcke, Tobias Hartung, Karl Jansen, Stefan Kühn
- **Posted on arXiv:** 01 February 2023
- **Published in PoS:** 01 February 2023

3.5.8.8 Quantum data learning for quantum simulations in high-energy physics [151]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

3.5.8.9 Simulating quantum field theories on continuous-variable quantum computers [166]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 15 March 2024
- **Published in Phys. Rev. A 110 (2024) 1, 012607:** 08 July 2024

3.5.8.10 Real-time scattering processes with continuous-variable quantum computers [167]

- **Authors:** Steven Abel, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 03 February 2025
- **Published in Phys. Rev. A 112 (2025), 012614:** 16 July 2025

3.6 Quantum Algorithms - Quantum Walks

3.6.1 Event Generation

3.6.1.1 Quantum walk approach to simulating parton showers [144]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

3.7 Quantum Annealing

3.7.1 Anomaly Detection

3.7.1.1 A Quantum Algorithm for Model-Independent Searches for New Physics [23]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

3.7.2 Detector Technologies and Simulations

3.7.2.1 Unfolding measurement distributions via quantum annealing [111]

- **Authors:** Kyle Cormier, Riccardo Di Sipio, Peter Wittek
- **Posted on arXiv:** 22 August 2019
- **Published in JHEP:** 22 November 2019

3.7.3 Event Classification

3.7.3.1 Solving a Higgs optimization problem with quantum annealing for machine learning [122]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

3.7.3.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [123]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

3.7.3.3 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [124]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

3.7.3.4 Quantum algorithm for the classification of supersymmetric top quark events [34]

- **Authors:** Pedrame Bargassa, Timothèe Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothèe Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

3.7.3.5 Completely quantum neural networks [125]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

3.7.4 Jet Reconstruction

3.7.4.1 Adiabatic quantum algorithm for multijet clustering in high energy physics [156]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

3.7.4.2 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [157]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

3.7.4.3 Quantum annealing for jet clustering with thrust [158]

- **Authors:** Andrea Delgado, Jesse Thaler
- **Posted on arXiv:** 05 May 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.7.4.4 Degeneracy engineering for classical and quantum annealing: A case study of sparse linear regression in collider physics [159]

- **Authors:** Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, Jesse Thaler
- **Posted on arXiv:** 20 May 2022
- **Published in Phys. Rev. D 106, 056008 (2022):** 01 September 2022

3.7.5 Lattice Field Theories - Gauge Theories

3.7.5.1 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [198]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

3.7.5.2 SU(2) lattice gauge theory on a quantum annealer [199]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

3.7.6 Track Reconstruction

3.7.6.1 A pattern recognition algorithm for quantum annealers [230]

- **Authors:** Frèdèric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
- **Published in Comput.Softw.Big Sci.:** 09 December 2019

3.7.6.2 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [231]

- **Authors:** Souvik Das, Andrew J. Wildridge, Andreas Jung
- **Posted on arXiv:** 21 March 2019

3.7.6.3 Charged particle tracking with quantum annealing optimization [232]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

3.7.6.4 Quantum annealing algorithms for track pattern recognition [233]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

3.7.6.5 Particle track classification using quantum associative memory [234]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

3.7.6.6 Quantum-Annealing-Inspired Algorithms for Track Reconstruction at High-Energy Colliders [235]

- **Authors:** Hideki Okawa, Qing-Guo Zeng, Xian-Zhe Tao, Man-Hong Yung
- **Posted on arXiv:** 22 February 2024
- **Published in Comput.Softw.Big Sci.:** 28 August 2024

3.7.7 Crosslists

3.7.7.1 Quantum Algorithms for Jet Clustering [155]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.7.7.2 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.7.7.3 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.7.7.4 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi
- **Published in Front.in Phys.:** 27 May 2022

3.7.7.5 Preparations for quantum simulations of quantum chromodynamics in <math display=”inline”><mrow><mn>1</mn><mo>+</mo><mn>1</mn></mrow></ dimensions. I. Axial gauge [189]

- **Authors:** Roland C. Farrell, Ivan A. Chernyshev, Sarah J.M. Powell, Nikita A. Zemlevskiy, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 04 July 2022
- **Published in Phys. Rev. D 107, 054512 (2023):** 01 March 2023

3.7.7.6 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.8 Quantum Entanglement and Bell Inequalities

3.8.1 Beyond the Standard Model

3.8.1.1 Flavor patterns of fundamental particles from quantum entanglement? [29]

- **Authors:** Jesse Thaler, Sokratis Trifinopoulos
- **Posted on arXiv:** 30 October 2024
- **Published in Phys.Rev.D:** 01 March 2025

3.8.2 Effective Field Theories

3.8.2.1 Minimal entanglement and emergent symmetries in low-energy QCD [118]

- **Authors:** Qiaofeng Liu, Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 October 2022
- **Published in Phys.Rev.C 107, 025204 (2023):** 14 February 2023

3.8.3 Quantum Chromodynamics

3.8.3.1 Symmetry from entanglement suppression [211]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

3.8.4 Quantum Correlations at Colliders

3.8.4.1 Entanglement and quantum tomography with top quarks at the LHC [212]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

3.8.4.2 Testing Bell Inequalities at the LHC with Top-Quark Pairs [213]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

3.8.4.3 Quantum tops at the LHC: from entanglement to Bell inequalities [214]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

3.8.4.4 Quantum SMEFT tomography: Top quark pair production at the LHC [119]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

3.8.4.5 Quantum information with top quarks in QCD [215]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

3.8.4.6 Improved tests of entanglement and Bell inequalities with LHC tops [216]

- **Authors:** J.A. Aguilar-Saavedra, J.A. Casas
- **Posted on arXiv:** 01 May 2022
- **Published in Eur.Phys.J.C:** 03 August 2022

3.8.4.7 Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs [217]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli
- **Posted on arXiv:** 24 August 2022
- **Published in Eur.Phys.J.C:** 19 February 2023

3.8.4.8 Quantum Discord and Steering in Top Quarks at the LHC [218]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 08 September 2022
- **Published in Phys. Rev. Lett. 130, 221801 (2023):** 31 May 2023

3.8.4.9 Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders [219]

- **Authors:** Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 01 February 2023
- **Published in Eur.Phys.J.C:** 15 September 2023

3.8.4.10 Three-Body Entanglement in Particle Decays [220]

- **Authors:** Kazuki Sakurai, Michael Spannowsky
- **Posted on arXiv:** 02 October 2023
- **Published in Phys.Rev.Lett.:** 09 April 2024

3.8.4.11 Can Bell inequalities be tested via scattering cross-section at colliders ? [221]

- **Authors:** Song Li, Wei Shen, Jin Min Yang
- **Posted on arXiv:** 02 January 2024
- **Published in Eur.Phys.J.C:** 21 November 2024

3.8.4.12 Quantum detection of new physics in top-quark pair production at the LHC [222]

- **Authors:** Fabio Maltoni, Claudio Severi, Simone Tentori, Eleni Vryonidou
- **Posted on arXiv:** 16 January 2024
- **Published in JHEP:** 15 March 2024

3.8.4.13 Full quantum tomography of top quark decays [223]

- **Authors:** J.A. Aguilar-Saavedra
- **Posted on arXiv:** 22 February 2024
- **Published in Phys.Lett.B:** 04 July 2024

3.8.4.14 Quantum information meets high-energy physics: input to the update of the European strategy for particle physics [224]

- **Authors:** Yoav Afik, Federica Fabbri, Matthew Low, Luca Marzola, Juan Antonio Aguilar-Saavedra, Mohammad Mahdi Altakach, Nedaa Alexandra Asbah, Yang Bai, Hannah Banks, Alan J. Barr, Alexander Bernal, Thomas E. Browder, Paweł Caban, J. Alberto Casas, Kun Cheng, Frèdèric Dèliot, Regina Demina, Antonio Di Domenico, Michał Eckstein, Marco Fabbrichesi, Benjamin Fuks, Emidio Gabrielli, Dorival Gonçalves, Radosław Grabarczyk, Michele Grossi, Tao Han, Timothy J. Hobbs, Paweł Horodecki, James Howarth, Shih-Chieh Hsu, Stephen Jiggins, Eleanor Jones, Andreas W. Jung, Andrea Helen Knue, Steffen Korn, Theodota Lagouri, Priyanka Lamba, Gabriel T. Landi, Haifeng Li, Qiang Li, Ian Low, Fabio Maltoni, Josh McFayden, Navin McGinnis, Roberto A. Morales, Jesús M. Moreno, Juan Ramón Muñoz de Nova, Giulia Negro, Davide Pagani, Giovanni Pelliccioli, Michele Pinamonti, Laura Pintucci, Baptiste Ravina, Alim Ruzi, Kazuki Sakurai, Ethan Simpson, Maximiliano Sioli, Shufang Su, Sokratis Trifinopoulos, Sven E. Vahsen, Sofia Vallecorsa, Alessandro Vicini, Marcel Vos, Eleni Vryonidou, Chris D. White, Martin J. White, Andrew J. Wildridge, Tong Arthur Wu, Laura Zani, Yulei Zhang, Knut Zoch
- **Posted on arXiv:** 31 March 2025
- **Published in Eur.Phys.J.Plus:** 09 September 2025

3.8.4.15 Colliders are Testing neither Locality via Bell's Inequality nor Entanglement versus Non-Entanglement [225]

- **Authors:** Steven A. Abel, Herbi K. Dreiner, Rhitaja Sengupta, Lorenzo Ubaldi
- **Posted on arXiv:** 21 July 2025

3.8.5 Crosslists

3.8.5.1 Testing Bell Inequalities at the LHC with Top-Quark Pairs [213]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

3.8.5.2 Snowmass white paper: Quantum information in quantum field theory and quantum gravity [11]

- **Authors:** Thomas Faulkner, Thomas Hartman, Matthew Headrick, Mukund Rangamani, Brian Swingle
- **Posted on arXiv:** 14 March 2022

3.8.5.3 Quantum entanglement and Bell inequality violation at colliders [7]

- **Authors:** Alan J. Barr, Marco Fabbrichesi, Roberto Floreanini, Emidio Gabrielli, Luca Marzola
- **Posted on arXiv:** 12 February 2024
- **Published in Prog.Part.Nucl.Phys.:** 19 July 2024

3.9 Quantum Machine Learning - Supervised Methods

3.9.1 Anomaly Detection

3.9.1.1 Unravelling physics beyond the standard model with classical and quantum anomaly detection [24]

- **Authors:** Julian Schuhmacher, Laura Boggia, Vasilis Belis, Ema Puljak, Michele Grossi, Maurizio Pierini, Sofia Vallecorsa, Francesco Tacchino, Panagiotis Barkoutsos, Ivano Tavernelli
- **Posted on arXiv:** 25 January 2023
- **Published in Mach.Learn.Sci.Tech.:** 16 November 2023

3.9.2 Event Classification

3.9.2.1 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [126]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevior
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

3.9.2.2 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [127]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

3.9.2.3 Higgs analysis with quantum classifiers [128]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

3.9.2.4 Enforcing exact permutation and rotational symmetries in the application of quantum neural networks on point cloud datasets [129]

- **Authors:** Zhelun Li, Lento Nagano, Koji Terashi
- **Posted on arXiv:** 17 May 2024
- **Published in Phys.Rev.Res.:** 10 October 2024

3.9.2.5 From Reachability to Learnability: Geometric Design Principles for Quantum Neural Networks [130]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky
- **Posted on arXiv:** 03 March 2026

3.9.3 Jet Reconstruction

3.9.3.1 Quantum-inspired machine learning on high-energy physics data [160]

- **Authors:** Timo Felser, Marco Trenti, Lorenzo Sestini, Alessio Gianelle, Davide Zuliani, Donatella Lucchesi, Simone Montangero
- **Posted on arXiv:** 28 April 2020
- **Published in npj Quantum Inf.:** 15 July 2021

3.9.3.2 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [161]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

3.9.3.3 Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data [162]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 21 February 2022
- **Published in Phys. Rev. A 106 (Dec, 2022) 062423:** 19 December 2022

3.9.4 Lattice Field Theories - Gauge Theories

3.9.4.1 Quantum data learning for quantum simulations in high-energy physics [151]

- **Authors:** Lento Nagano, Alexander Miessen, Tamiya Onodera, Ivano Tavernelli, Francesco Tacchino, Koji Terashi
- **Posted on arXiv:** 29 June 2023
- **Published in Phys.Rev.Res.:** 15 December 2023

3.9.5 Track Reconstruction

3.9.5.1 Reconstructing charged particle track segments with a quantum-enhanced support vector machine [236]

- **Authors:** Philippa Duckett, Gabriel Facini, Marcin Jastrzebski, Sarah Malik, Tim Scanlon, Sebastien Rettie
- **Posted on arXiv:** 14 December 2022
- **Published in Phys.Rev.D:** 01 March 2024

3.9.6 Crosslists

3.9.6.1 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.9.6.2 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico

Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023
- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.9.6.3 Quantum-centric Supercomputing for Physics Research [116]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.10 Quantum Machine Learning - Unsupervised Methods

3.10.1 Anomaly Detection

3.10.1.1 Quantum anomaly detection in the latent space of proton collision events at the LHC [25]

- **Authors:** Vasilis Belis, Kinga Anna Woźniak, Ema Puljak, Panagiotis Barkoutsos, Günther Dissertori, Michele Grossi, Maurizio Pierini, Florentin Reiter, Ivano Tavernelli, Sofia Vallecorsa
- **Posted on arXiv:** 25 January 2023
- **Published in Commun.Phys.:** 14 October 2024

3.10.2 Detector Technologies and Simulations

3.10.2.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [112]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.10.2.2 Running the Dual-PQC GAN on noisy simulators and real quantum hardware [113]

- **Authors:** Su Yeon Chang, Edwin Agnew, Elías F. Combarro, Michele Grossi, Steven Herbert, Sofia Vallecorsa
- **Posted on arXiv:** 30 May 2022
- **Published in J.Phys.Conf.Ser.:** 2023

3.10.2.3 Precise image generation on current noisy quantum computing devices [114]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borras, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

3.10.3 Event Classification

3.10.3.1 Guided graph compression for quantum graph neural networks [131]

- **Authors:** Mikel Casals, Vasilis Belis, Elias F. Combarro, Eduard Alarcón, Sofia Vallecorsa, Michele Grossi
- **Posted on arXiv:** 11 June 2025
- **Published in Mach.Learn.Sci.Tech.:** 12 September 2025

3.10.4 Event Generation

3.10.4.1 Style-based quantum generative adversarial networks for Monte Carlo events [145]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

3.10.4.2 Quantum integration of elementary particle processes [146]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

3.10.4.3 Generative invertible quantum neural networks [147]

- **Authors:** Armand Rousselot, Michael Spannowsky
- **Posted on arXiv:** 24 February 2023
- **Published in SciPost Phys. 16, 146 (2024):** 04 June 2024

3.10.4.4 Quantum-centric Supercomputing for Physics Research [116]

- **Authors:** Vincent R. Pascuzzi, Antonio Córcoles
- **Posted on arXiv:** 21 August 2024

3.10.5 Jet Reconstruction

3.10.5.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [163]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

3.10.5.2 Quantum clustering and jet reconstruction at the LHC [164]

- **Authors:** Jorge J. Martínez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 13 April 2022
- **Published in Physical Review D 106, 036021 (2022):** 01 August 2022

3.10.6 Crosslists

3.10.6.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [110]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.10.6.2 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo
- **Posted on arXiv:** 15 March 2022

3.10.6.3 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023

- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.10.6.4 Machine learning for anomaly detection in particle physics [6]

- **Authors:** Vasilis Belis, Patrick Odagiu, Thea Klaeboe Aarrestad

- **Posted on arXiv:** 20 December 2023

- **Published in Reviews in Physics, vol. 12, 2024, 100091:** 18 January 2024

3.11 Quantum Machine Learning - Variational Quantum Algorithms

3.11.1 Anomaly Detection

3.11.1.1 Anomaly detection in high-energy physics using a quantum autoencoder [26]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi

- **Posted on arXiv:** 09 December 2021

- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

3.11.1.2 Quantum anomaly detection for collider physics [27]

- **Authors:** Sulaiman Alvi, Christian W. Bauer, Benjamin Nachman

- **Posted on arXiv:** 16 June 2022

- **Published in JHEP:** 22 February 2023

3.11.2 Beyond the Standard Model

3.11.2.1 Hybrid Quantum-Classical Graph Convolutional Network [30]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

3.11.3 Event Classification

3.11.3.1 Event Classification with Quantum Machine Learning in High-Energy Physics [31]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

3.11.3.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [32]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

3.11.3.3 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [132]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

3.11.3.4 Hybrid actor-critic algorithm for quantum reinforcement learning at CERN beam lines [133]

- **Authors:** Michael Schenk, Elías F. Combarro, Michele Grossi, Verena Kain, Kevin Shing Bruce Li, Mircea-Marian Popa, Sofia Vallecorsa
- **Posted on arXiv:** 22 September 2022
- **Published in Quantum Sci.Technol.:** 21 February 2024

3.11.3.5 Fitting a collider in a quantum computer: tackling the challenges of quantum machine learning for big datasets [134]

- **Authors:** Miguel Caçador Peixoto, Nuno Filipe Castro, Miguel Crispim Romão, Maria Gabriela Jordão Oliveira, Inês Ochoa
- **Posted on arXiv:** 06 November 2022
- **Published in Front. Artif. Intell. 6 (2023) 1268852:** 20 November 2023

3.11.3.6 Quantum Metric Learning for New Physics Searches at the LHC [35]

- **Authors:** A. Hammad, Kyoungchul Kong, Myeonghun Park, Soyoung Shim
- **Posted on arXiv:** 28 November 2023

3.11.3.7 Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics [135]

- **Authors:** Eyup B. Unlu, Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva
- **Posted on arXiv:** 01 February 2024
- **Published in Axioms v. 13, no 3, (2024) 187:** 13 March 2024

3.11.3.8 Quantum Vision Transformers for Quark–Gluon Classification [136]

- **Authors:** Marçal Comajoan Cara, Gopal Ramesh Dahale, Zhongtian Dong, Roy T. Forestano, Sergei Gleyzer, Daniel Justice, Kyoungchul Kong, Tom Magorsch, Konstantin T. Matchev, Katia Matcheva, Eyup B. Unlu
- **Posted on arXiv:** 16 May 2024
- **Published in Axioms 2024, 13(5), 323:** 13 May 2024

3.11.4 Event Generation

3.11.4.1 Partonic collinear structure by quantum computing [148]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

3.11.4.2 Unsupervised quantum circuit learning in high energy physics [149]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.11.4.3 Quantum Wishlist: Lessons from Parton Showers [150]

- **Authors:** Masahito Yamazaki
- **Posted on arXiv:** 25 February 2025
- **Published in PoS:** 24 October 2025

3.11.5 Jet Reconstruction

3.11.5.1 Quantum Machine Learning for b-jet charge identification [165]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

3.11.6 Lattice Field Theories - Gauge Theories

3.11.6.1 SU(2) hadrons on a quantum computer via a variational approach [200]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

3.11.7 Lattice Field Theories - Schwinger Model

3.11.7.1 Quantum-classical computation of Schwinger model dynamics using quantum computers [206]

- **Authors:** N. Klco, E.F. Dumitrescu, A.J. McCaskey, T.D. Morris, R.C. Pooser, M. Sanz, E. Solano, P. Lougovski, M.J. Savage
- **Posted on arXiv:** 08 March 2018
- **Published in Phys. Rev. A 98, 032331 (2018):** 29 September 2018

3.11.7.2 Quantum-Classical Simulation of Quantum Field Theory by Quantum Circuit Learning [207]

- **Authors:** Kazuki Ikeda
- **Posted on arXiv:** 27 November 2023
- **Published in Annalen Phys.:** 01 June 2025

3.11.8 Track Reconstruction

3.11.8.1 Quantum convolutional neural networks for high energy physics data analysis [33]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

3.11.8.2 Hybrid quantum classical graph neural networks for particle track reconstruction [237]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

3.11.8.3 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [238]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in J.Phys.Conf.Ser.:** 2023

3.11.8.4 Particle track reconstruction with noisy intermediate-scale quantum computers [239]

- **Authors:** Tim Schwägerl, Cigdem Issever, Karl Jansen, Teng Jian Khoo, Stefan Kühn, Cenk Tüysüz, Hannsjörg Weber
- **Posted on arXiv:** 23 March 2023
- **Published in J.Phys.Conf.Ser.:** 2026

3.11.8.5 Charged particle reconstruction for future high energy colliders with Quantum Approximate Optimization Algorithm [240]

- **Authors:** Hideki Okawa
- **Posted on arXiv:** 16 October 2023

3.11.9 Crosslists

3.11.9.1 Simulating Lattice Gauge Theories within Quantum Technologies [181]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.11.9.2 Quantum Machine Learning in High Energy Physics [3]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.11.9.3 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [112]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.11.9.4 Higgs analysis with quantum classifiers [128]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

3.11.9.5 Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [13]

- **Authors:** Travis S. Humble, Andrea Delgado, Raphael Pooser, Christopher Seck, Ryan Bennink, Vicente Leyton-Ortega, C.-C. Joseph Wang, Eugene Dumitrescu, Titus Morris, Kathleen Hamilton, Dmitry Lyakh, Prasanna Date, Yan Wang, Nicholas A. Peters, Katherine J. Evans, Marcel Demarteau, Alex McCaskey, Thien Nguyen, Susan Clark, Melissa Reville, Alberto Di Meglio, Michele Grossi, Sofia Vallecorsa, Kerstin Borras, Karl Jansen, Dirk Krücker

- **Posted on arXiv:** 14 March 2022

3.11.9.6 Quantum Computing for Data Analysis in High-Energy Physics [14]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton, Prasanna Date, Jean-Roch Vlimant, Duarte Magano, Yasser Omar, Pedrame Bargassa, Anthony Francis, Alessio Gianelle, Lorenzo Sestini, Donatella Lucchesi, Davide Zuliani, Davide Nicotra, Jacco de Vries, Dominica Dibenedetto, Miriam Lucio Martinez, Eduardo Rodrigues, Carlos Vazquez Sierra, Sofia Vallecorsa, Jesse Thaler, Carlos Bravo-Prieto, su Yeon Chang, Jeffrey Lazar, Carlos A. Argüelles, Jorge J. Martinez de Lejarza, Leandro Cieri, Germán Rodrigo

- **Posted on arXiv:** 15 March 2022

3.11.9.7 Quantum Computing Applications in Future Colliders [4]

- **Authors:** Heather M. Gray, Koji Terashi

- **Published in Front.in Phys.:** 27 May 2022

3.11.9.8 Quantum Computing for High-Energy Physics: State of the Art and Challenges [20]

- **Authors:** Alberto Di Meglio, Karl Jansen, Ivano Tavernelli, Constantia Alexandrou, Srinivasan Arunachalam, Christian W. Bauer, Kerstin Borras, Stefano Carrazza, Arianna Crippa, Vincent Croft, Roland de Putter, Andrea Delgado, Vedran Dunjko, Daniel J. Egger, Elias Fernandez-Combarro, Elina Fuchs, Lena Funcke, Daniel Gonzalez-Cuadra, Michele Grossi, Jad C. Halimeh, Zoe Holmes, Stefan Kuhn, Denis Lacroix, Randy Lewis, Donatella Lucchesi, Miriam Lucio Martinez, Federico Meloni, Antonio Mezzacapo, Simone Montangero, Lento Nagano, Vincent R. Pascuzzi, Voica Radescu, Enrique Rico Ortega, Alessandro Roggero, Julian Schuhmacher, Joao Seixas, Pietro Silvi, Panagiotis Spentzouris, Francesco Tacchino, Kristan Temme, Koji Terashi, Jordi Tura, Cenk Tuysuz, Sofia Vallecorsa, Uwe-Jens Wiese, Shinjae Yoo, Jinglei Zhang

- **Posted on arXiv:** 06 July 2023

- **Published in PRX Quantum 5, 037001 (2024):** 01 August 2024

3.11.9.9 Precise image generation on current noisy quantum computing devices [114]

- **Authors:** Florian Rehm, Sofia Vallecorsa, Kerstin Borras, Dirk Krücker, Michele Grossi, Valle Varo
- **Posted on arXiv:** 11 July 2023
- **Published in IOP Quantum Science and Technology (October 2023):** 30 October 2023

3.11.9.10 Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits [202]

- **Authors:** Roland C. Farrell, Marc Illa, Anthony N. Ciavarella, Martin J. Savage
- **Posted on arXiv:** 08 August 2023
- **Published in PRX Quantum 5 (2024) 2, 020315:** 01 April 2024

3.11.9.11 Quantum Algorithms in Particle Physics [152]

- **Authors:** Germán Rodrigo
- **Posted on arXiv:** 29 January 2024
- **Published in Acta Phys.Polon.Supp.:** 2024

3.11.9.12 Scattering wave packets of hadrons in gauge theories: Preparation on a quantum computer [195]

- **Authors:** Zohreh Davoudi, Chung-Chun Hsieh, Saurabh V. Kadam
- **Posted on arXiv:** 01 February 2024
- **Published in Quantum:** 11 November 2024

3.11.9.13 Steps toward quantum simulations of hadronization and energy loss in dense matter [203]

- **Authors:** Roland C. Farrell, Marc Illa, Martin J. Savage
- **Posted on arXiv:** 10 May 2024
- **Published in Phys.Rev.C 111 (2025) 1, 015202:** 14 January 2025

3.12 Quantum Sensors - Atomic/Molecular/Nuclear Sensors

3.12.1 Dark Matter - Particle-like Dark Matter

3.12.1.1 Trapped Electrons and Ions as Particle Detectors [37]

- **Authors:** Daniel Carney, Hartmut Häffner, David C. Moore, Jacob M. Taylor
- **Posted on arXiv:** 12 April 2021
- **Published in Phys Rev Lett, Vol. 127, No. 6, 2021. "Editor's suggestion":** 05 August 2021

3.12.1.2 Millicharged Dark Matter Detection with Ion Traps [38]

- **Authors:** Dmitry Budker, Peter W. Graham, Harikrishnan Ramani, Ferdinand Schmidt-Kaler, Christian Smorra, Stefan Ulmer
- **Posted on arXiv:** 11 August 2021
- **Published in PRX Quantum:** 01 February 2022

3.12.2 Dark Matter - Wave-like Dark Matter

3.12.2.1 Intensity interferometry for ultralight bosonic dark matter detection [55]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball
- **Posted on arXiv:** 05 February 2022
- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

3.12.2.2 One-Electron Quantum Cyclotron as a Milli-eV Dark-Photon Detector [56]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Roni Harnik, Thomas G. Myers, Harikrishnan Ramani, Benedict A.D. Sukra, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 12 August 2022
- **Published in Phys. Rev. Lett. 129, 261801 (2022):** 23 December 2022

3.12.2.3 Detection of bosenovae with quantum sensors on Earth and in space [57]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

3.12.2.4 Quantum entanglement of ions for light dark matter detection [58]

- **Authors:** Asuka Ito, Ryuichiro Kitano, Wakutaka Nakano, Ryoto Takai
- **Posted on arXiv:** 20 November 2023
- **Published in JHEP:** 16 February 2024

3.12.2.5 A Nuclear Interferometer for Ultra-Light Dark Matter Detection [59]

- **Authors:** Hannah Banks, Elina Fuchs, Matthew McCullough
- **Posted on arXiv:** 15 July 2024

3.12.2.6 Highly excited electron cyclotron for QCD axion and dark-photon detection [60]

- **Authors:** Xing Fan, Gerald Gabrielse, Peter W. Graham, Harikrishnan Ramani, Samuel S.Y. Wong, Yawen Xiao
- **Posted on arXiv:** 07 October 2024
- **Published in Phys.Rev.D 111 (2025) 7, 075022:** 01 April 2025

3.12.2.7 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [61]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

3.12.2.8 Sensitively searching for microwave dark photons with atomic ensembles [62]

- **Authors:** Suirong He, De He, Yufen Li, Li Gao, Xianing Feng, Hao Zheng, L.F. Wei
- **Posted on arXiv:** 01 December 2024
- **Published in Phys.Rev.D:** 15 September 2025

3.12.2.9 Searches for exotic spin-dependent interactions with spin sensors [63]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

3.12.2.10 Spin squeezing of macroscopic nuclear spin ensembles [64]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

3.12.2.11 Detecting Dark Matter Using Optically Trapped Rydberg Atom Tweezer Arrays [65]

- **Authors:** So Chigusa, Taiyo Kasamaki, Toshi Kusano, Takeo Moroi, Kazunori Nakayama, Naoya Ozawa, Yoshiro Takahashi, Atsuhiro Umemoto, Amar Vutha
- **Posted on arXiv:** 17 July 2025
- **Published in Phys. Rev. Lett. 136, 151801 (2026):** 14 April 2026

3.12.2.12 Superradiant interactions for relic detection with entangled nuclear spins [66]

- **Authors:** Marios Galanis, Onur Hosten, Asimina Arvanitaki, Savas Dimopoulos
- **Posted on arXiv:** 28 August 2025
- **Published in Phys.Rev.A:** 10 June 2026

3.12.2.13 On the Speed-up of Wave-like Dark Matter Searches with Entangled Qubits [67]

- **Authors:** Arushi Bodas, Sohitri Ghosh, Roni Harnik
- **Posted on arXiv:** 13 October 2025

3.12.2.14 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [68]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

3.12.3 Detector Technologies and Simulations

3.12.3.1 Quantum Systems for Enhanced High Energy Particle Physics Detectors [115]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov
- **Published in nan:** 24 June 2022

3.12.3.2 Quantum sensing for particle physics [108]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.12.4 Crosslists

3.12.4.1 Dark Matter Direct Detection with Accelerometers [84]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

3.12.4.2 Search for New Physics with Atoms and Molecules [2]

- **Authors:** M.S. Safronova, D. Budker, D. DeMille, Derek F. Jackson Kimball, A. Derevianko, C.W. Clark
- **Posted on arXiv:** 04 October 2017
- **Published in Reviews of Modern Physics, 90, 025008 (2018):** 30 June 2018

3.12.4.3 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.12.4.4 Simulating Lattice Gauge Theories within Quantum Technologies [181]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller

- **Posted on arXiv:** 31 October 2019

- **Published in Eur.Phys.J.D:** 04 August 2020

3.12.4.5 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova

- **Posted on arXiv:** 14 March 2022

3.12.4.6 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J.

Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.12.4.7 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.12.4.8 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage

- **Posted on arXiv:** 14 September 2022

3.12.4.9 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

3.12.4.10 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [76]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis

- **Posted on arXiv:** 07 August 2024

- **Published in Phys.Rev.D:** 01 March 2025

3.12.4.11 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [109]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

3.12.4.12 Quantum Error Correction-like Noise Mitigation for Wave-like Dark Matter Searches with Quantum Sensors [82]

- **Authors:** Hajime Fukuda, Takeo Moroi, Thanaporn Sichanugrist
- **Posted on arXiv:** 05 November 2025

3.13 Quantum Sensors - Optical/Photonic Sensors

3.13.1 Dark Matter - Wave-like Dark Matter

3.13.1.1 Laser Interferometers as Dark Matter Detectors [69]

- **Authors:** Evan D. Hall, Rana X. Adhikari, Valery V. Frolov, Holger Müller, Maxim Pospelov, Rana X Adhikari
- **Posted on arXiv:** 03 May 2016
- **Published in Phys.Rev.D:** 23 October 2018

3.13.1.2 Accelerating dark-matter axion searches with quantum measurement technology [70]

- **Authors:** Huaixiu Zheng, Matti Silveri, R.T. Brierley, S.M. Girvin, K.W. Lehnert
- **Posted on arXiv:** 08 July 2016

3.13.1.3 Searching for Dark Matter with a Superconducting Qubit [71]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

3.13.1.4 Detecting Hidden Photon Dark Matter Using the Direct Excitation of Transmon Qubits [72]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 07 December 2022
- **Published in Phys.Rev.Lett.:** 22 November 2023

3.13.1.5 Quantum measurements in fundamental physics: a user's manual [73]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023

3.13.1.6 Quantum Enhancement in Dark Matter Detection with Quantum Computation [74]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 17 November 2023
- **Published in Phys.Rev.Lett.:** 08 July 2024

3.13.1.7 Search for QCD axion dark matter with transmon qubits and quantum circuit [75]

- **Authors:** Shion Chen, Hajime Fukuda, Toshiaki Inada, Takeo Moroi, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 29 July 2024
- **Published in Phys.Rev.D:** 01 December 2024

3.13.1.8 Superradiant interactions of the cosmic neutrino background, axions, dark matter, and reactor neutrinos [76]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Marios Galanis
- **Posted on arXiv:** 07 August 2024
- **Published in Phys.Rev.D:** 01 March 2025

3.13.1.9 Axion bounds from quantum technology [77]

- **Authors:** Martin Bauer, Sreemanti Chakraborti, Guillaume Rostagni
- **Posted on arXiv:** 12 August 2024
- **Published in JHEP:** 05 May 2025

3.13.1.10 Quantum-enhanced dark matter detection with in-cavity control: mitigating the Rayleigh curse [78]

- **Authors:** Haowei Shi, Anthony J. Brady, Wojciech Górecki, Lorenzo Maccone, Roberto Di Candia, Quntao Zhuang
- **Posted on arXiv:** 06 September 2024
- **Published in npj Quantum Information 11, 48 (2025):** 17 March 2025

3.13.1.11 Transmon qubit modeling and characterization for Dark Matter search [79]

- **Authors:** Roberto Moretti, Danilo Labranca, Pietro Campana, Rodolfo Carobene, Marco Gobbo, Manuel A. Castellanos-Beltran, David Olaya, Peter F. Hopkins, Leonardo Banchi, Matteo Borghesi, Alessandro Candido, Stefano Carrazza, Hervè Atsè Corti, Alessandro D'Elia, Marco Faverzani, Elena Ferri, Angelo Nucciotti, Luca Origo, Andrea Pasquale, Alex Stephane Piedjou Komnang, Alessio Rettaroli, Simone Tocci, Claudio Gatti, Andrea Giachero
- **Posted on arXiv:** 09 September 2024
- **Published in IEEE Trans.Quantum Eng.:** 2026

3.13.1.12 Flux-Tunable Cavity for Dark Matter Detection [80]

- **Authors:** Fang Zhao, Ziqian Li, Akash V. Dixit, Tanay Roy, Andrei Vrajitoarea, Riju Banerjee, Alexander Anferov, Kan-Heng Lee, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 12 January 2025
- **Published in Phys.Rev.Lett.:** 13 November 2025

3.13.1.13 Quantum Enhanced Dark-Matter Search with Entangled Fock States in High-Quality Cavities [81]

- **Authors:** Benjamin Freiman, Xinyuan You, Andy C.Y. Li, Raphael Cervantes, Taeyoon Kim, Anna Grasselino, Roni Harnik, Yao Lu
- **Posted on arXiv:** 30 October 2025

3.13.1.14 Quantum Error Correction-like Noise Mitigation for Wave-like Dark Matter Searches with Quantum Sensors [82]

- **Authors:** Hajime Fukuda, Takeo Moroi, Thanaporn Sichanugrist
- **Posted on arXiv:** 05 November 2025

3.13.2 Gravitational Waves

3.13.2.1 Quantum sensor networks as exotic field telescopes for multi-messenger astronomy [106]

- **Authors:** Conner Dailey, Colin Bradley, Derek F. Jackson Kimball, Ibrahim A. Sulai, Szymon Pustelny, Arne Wickenbrock, Andrei Derevianko
- **Posted on arXiv:** 11 February 2020
- **Published in Nature Astron.:** 02 November 2020

3.13.2.2 Electromagnetic cavities as mechanical bars for gravitational waves [107]

- **Authors:** Asher Berlin, Diego Blas, Raffaele Tito D'Agnolo, Sebastian A.R. Ellis, Roni Harnik, Yonatan Kahn, Jan Schütte-Engel, Michael Wentzel
- **Posted on arXiv:** 02 March 2023
- **Published in Phys.Rev.D:** 15 October 2023

3.13.2.3 Multimessenger astronomy beyond the Standard Model: New window from quantum sensors [109]

- **Authors:** Jason Arakawa, Muhammad H. Zaheer, Volodymyr Takhistov, Marianna S. Safronova, Joshua Eby, Charles Cheung
- **Posted on arXiv:** 12 February 2025
- **Published in JCAP:** 10 February 2026

3.13.3 Crosslists

3.13.3.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul

Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.13.3.2 Simulating Lattice Gauge Theories within Quantum Technologies [181]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.13.3.3 Intensity interferometry for ultralight bosonic dark matter detection [55]

- **Authors:** Hector Masia-Roig, Nataniel L. Figueroa, Ariday Bordon, Joseph A. Smiga, Yevgeny V. Stadnik, Dmitry Budker, Gary P. Centers, Alexander V. Gramolin, Paul S. Hamilton, Sami Khamis, Christopher A. Palm, Szymon Pustelny, Alexander O. Sushkov, Arne Wickenbrock, Derek F. Jackson Kimball
- **Posted on arXiv:** 05 February 2022
- **Published in Phys. Rev. D 108, 015003 (2023):** 01 July 2023

3.13.3.4 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.13.3.5 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.13.3.6 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.13.3.7 Quantum computing hardware for HEP algorithms and sensing [18]

- **Authors:** M. Sohaib Alam, Sergey Belomestnykh, Nicholas Bornman, Gustavo Cancelo, Yu-Chiu Chao, Mattia Checchin, Vinh San Dinh, Anna Grassellino, Erik J. Gustafson, Roni Harnik, Corey Rae Harrington McRae, Ziwen Huang, Keshav Kapoor, Taeyoon Kim, James B. Kowalkowski, Matthew J. Kramer, Yulia Krasnikova, Prem Kumar, Doga Murat Kurkcuoglu, Henry Lamm, Adam L. Lyon, Despina Milathianaki, Akshay Murthy, Josh Mutus, Ivan Nekrashevich, JinSu Oh, A. BarışÖzgüler, Gabriel Nathan Perdue, Matthew Reagor, Alexander Romanenko, James A. Sauls, Leandro Stefanazzi, Norm M. Tubman, Davide Venturelli, Changqing Wang, Xinyuan You, David M.T. van Zanten, Lin Zhou, Shaojiang Zhu, Silvia Zorzetti

- **Posted on arXiv:** 18 April 2022

3.13.3.8 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage
- **Posted on arXiv:** 14 September 2022

3.13.3.9 Quantum sensing for particle physics [108]

- **Authors:** Steven D. Bass, Michael Doser
- **Posted on arXiv:** 19 May 2023
- **Published in Nature Rev.Phys.:** 16 April 2024

3.13.3.10 Detection of bosenovae with quantum sensors on Earth and in space [57]

- **Authors:** Jason Arakawa, Joshua Eby, Marianna S. Safronova, Volodymyr Takhistov, Muhammad H. Zaheer
- **Posted on arXiv:** 28 June 2023
- **Published in Phys.Rev.D:** 01 October 2024

3.13.3.11 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek
- **Posted on arXiv:** 03 November 2023

3.13.3.12 Eliminating Incoherent Noise: A Coherent Quantum Approach in Multi-Sensor Dark Matter Detection [61]

- **Authors:** Jing Shu, Bin Xu, Yuan Xu
- **Posted on arXiv:** 29 October 2024

3.13.3.13 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [68]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

3.14 Quantum Sensors - Optomechanical Sensors

3.14.1 Dark Matter - Particle-like Dark Matter

3.14.1.1 Search for Kilogram-scale Dark Matter with Precision Displacement Sensors [39]

- **Authors:** Akio Kawasaki
- **Posted on arXiv:** 26 August 2018
- **Published in Phys. Rev. D 99, 023005 (2019):** 07 January 2019

3.14.1.2 Proposal for gravitational direct detection of dark matter [40]

- **Authors:** Daniel Carney, Sohitri Ghosh, Gordan Krnjaic, Jacob M. Taylor
- **Posted on arXiv:** 01 March 2019
- **Published in Phys.Rev.D:** 14 October 2020

3.14.1.3 Search for composite dark matter with optically levitated sensors [41]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

3.14.1.4 Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors [42]

- **Authors:** Gadi Afek, Daniel Carney, David C. Moore
- **Posted on arXiv:** 05 November 2021
- **Published in Phys.Rev.Lett.:** 09 March 2022

3.14.1.5 Models of ultraheavy dark matter visible to macroscopic mechanical sensing arrays [43]

- **Authors:** Carlos Blanco, Bahaa Elshimy, Rafael F. Lang, Robert Orlando
- **Posted on arXiv:** 29 December 2021
- **Published in Phys.Rev.D:** 01 June 2022

3.14.1.6 Requirements on quantum superpositions of macro-objects for sensing neutrinos [44]

- **Authors:** Eva Kilian, Marko Toroš, Frank F. Deppisch, Ruben Saakyan, Sougato Bose
- **Posted on arXiv:** 27 April 2022
- **Published in Phys. Rev. Research 5, 023012 (2023):** 07 April 2023

3.14.1.7 Searches for Massive Neutrinos with Mechanical Quantum Sensors [45]

- **Authors:** Daniel Carney, Kyle G. Leach, David C. Moore
- **Posted on arXiv:** 12 July 2022
- **Published in Phys.Rev.X:** 01 February 2023

3.14.1.8 Optomechanical dark matter instrument for direct detection [46]

- **Authors:** Christopher G. Baker, Warwick P. Bowen, Peter Cox, Matthew J. Dolan, Maxim Goryachev, Glen Harris
- **Posted on arXiv:** 16 June 2023
- **Published in Phys. Rev. D 110, 043005 (2024):** 02 August 2024

3.14.1.9 Probing levitodynamics with multi-stochastic forces and the simple applications on the dark matter detection in optical levitation experiment [47]

- **Authors:** Xi Cheng, Ji-Heng Guo, Wenyu Wang, Bin Zhu
- **Posted on arXiv:** 07 December 2023
- **Published in Nucl. Phys. B 1010 (2025) 116780:** 19 December 2024

3.14.1.10 Dark Matter Searches with Levitated Sensors [48]

- **Authors:** Eva Kilian, Markus Rademacher, Jonathan M.H. Gosling, Julian H. Iacoponi, Fiona Alder, Marko Toroš, Antonio Pontin, Chamkaur Ghag, Sougato Bose, Tania S. Monteiro, P.F. Barker
- **Posted on arXiv:** 31 January 2024
- **Published in AVS Quantum Sci. 6, 030503 (2024):** 2024

3.14.1.11 Probing light particles with optically trapped sensors through nucleon scattering [49]

- **Authors:** Bhaskar Dutta, Dilip Kumar Ghosh, Sk Jeesun
- **Posted on arXiv:** 31 January 2025
- **Published in Phys.Rev.D:** 01 January 2026

3.14.1.12 Mechanical sensors for ultraheavy dark matter searches via long-range forces [50]

- **Authors:** Juehang Qin, Dorian W.P. Amaral, Sunil A. Bhave, Erqian Cai, Daniel Carney, Rafael F. Lang, Shengchao Li, Alberto M. Marino, Giacomo Marocco, Claire Marvinney, Jared R. Newton, Jacob M. Taylor, Christopher Tunnell
- **Posted on arXiv:** 14 March 2025
- **Published in Phys.Rev.D 112 (2025) 7, 072003:** 01 October 2025

3.14.2 Dark Matter - Wave-like Dark Matter

3.14.2.1 Sound of Dark Matter: Searching for Light Scalars with Resonant-Mass Detectors [83]

- **Authors:** Asimina Arvanitaki, Savas Dimopoulos, Ken Van Tilburg
- **Posted on arXiv:** 07 August 2015
- **Published in Phys.Rev.Lett.:** 22 January 2016

3.14.2.2 Dark Matter Direct Detection with Accelerometers [84]

- **Authors:** Peter W. Graham, David E. Kaplan, Jeremy Mardon, Surjeet Rajendran, William A. Terrano
- **Posted on arXiv:** 18 December 2015
- **Published in Phys.Rev.D:** 20 April 2016

3.14.2.3 Ultralight dark matter detection with mechanical quantum sensors [85]

- **Authors:** Daniel Carney, Anson Hook, Zhen Liu, Jacob M. Taylor, Yue Zhao
- **Posted on arXiv:** 13 August 2019
- **Published in New Journal of Physics, Volume 23, February 2021:** 02 March 2021

3.14.2.4 Searching for scalar dark matter with compact mechanical resonators [86]

- **Authors:** Jack Manley, Russell Stump, Dalziel Wilson, Daniel Grin, Swati Singh
- **Posted on arXiv:** 16 October 2019
- **Published in Phys. Rev. Lett. 124, 151301 (2020):** 17 April 2020

3.14.2.5 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [87]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

3.14.2.6 Searching for vector dark matter with an optomechanical accelerometer [88]

- **Authors:** Jack Manley, Mitul Dey Chowdhury, Daniel Grin, Swati Singh, Dalziel J. Wilson
- **Posted on arXiv:** 09 July 2020
- **Published in Phys. Rev. Lett. 126, 061301 (2021):** 11 February 2021

3.14.2.7 Entanglement-enhanced optomechanical sensor array with application to dark matter searches [89]

- **Authors:** Anthony J. Brady, Xin Chen, Kewen Xiao, Yi Xia, Jack Manley, Mitul Dey Chowdhury, Zhen Liu, Roni Harnik, Dalziel J. Wilson, Zheshen Zhang, Quntao Zhuang
- **Posted on arXiv:** 13 October 2022
- **Published in Commun.Phys.:** 01 September 2023

3.14.2.8 Axion detection with optomechanical cavities [90]

- **Authors:** Clara Murgui, Yikun Wang, Kathryn M. Zurek
- **Posted on arXiv:** 15 November 2022
- **Published in Phys.Lett.B:** 27 June 2024

3.14.2.9 Superfluid helium ultralight dark matter detector [91]

- **Authors:** M. Hirschel, V. Vadakkumbatt, N.P. Baker, F.M. Schweizer, J.C. Sankey, S. Singh, J.P. Davis
- **Posted on arXiv:** 14 September 2023
- **Published in Physical Review D 109, 095011 (2024):** 01 May 2024

3.14.2.10 Vector wave dark matter and terrestrial quantum sensors [92]

- **Authors:** Dorian W.P. Amaral, Mudit Jain, Mustafa A. Amin, Christopher Tunnell
- **Posted on arXiv:** 04 March 2024
- **Published in JCAP:** 21 June 2024

3.14.3 Gravitational Waves

3.14.3.1 Quantum interactions between a laser interferometer and gravitational waves [153]

- **Authors:** Belinda Pang, Yanbei Chen
- **Posted on arXiv:** 28 August 2018
- **Published in Phys. Rev. D 98, 124006 (2018):** 11 December 2018

3.14.3.2 Superconducting Levitated Detector of Gravitational Waves [154]

- **Authors:** Daniel Carney, Gerard Higgins, Giacomo Marocco, Michael Wentzel
- **Posted on arXiv:** 02 August 2024
- **Published in Carney (2025) Phys. Rev. Lett. 134, 181402:** 06 May 2025

3.14.4 Crosslists

3.14.4.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi
- **Posted on arXiv:** 29 March 2018

3.14.4.2 Mechanical quantum sensing in the search for dark matter [9]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao

- **Posted on arXiv:** 13 August 2020

- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

3.14.4.3 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova

- **Posted on arXiv:** 14 March 2022

3.14.4.4 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.14.4.5 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.14.4.6 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage

- **Posted on arXiv:** 14 September 2022

3.14.4.7 Quantum sensing for particle physics [108]

- **Authors:** Steven D. Bass, Michael Doser

- **Posted on arXiv:** 19 May 2023

- **Published in Nature Rev.Phys.:** 16 April 2024

3.14.4.8 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

3.14.4.9 Quantum measurements in fundamental physics: a user's manual [73]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco

- **Posted on arXiv:** 13 November 2023

3.14.4.10 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [68]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

3.15 Quantum Sensors - Solid State Sensors

3.15.1 Dark Matter - Particle-like Dark Matter

3.15.1.1 Dark matter direct detection with quantum dots [51]

- **Authors:** Carlos Blanco, Rouven Essig, Marivi Fernandez-Serra, Harikrishnan Ramani, Oren Slone
- **Posted on arXiv:** 10 August 2022
- **Published in Phys.Rev.D:** 01 May 2023

3.15.1.2 Dark Matter Induced Power in Quantum Devices [52]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 17 October 2022
- **Published in Phys. Rev. Lett. 132, 121801 (2024):** 22 March 2024

3.15.1.3 Transmon Qubit constraints on dark matter-nucleon scattering [53]

- **Authors:** Anirban Das, Noah Kurinsky, Rebecca K. Leane
- **Posted on arXiv:** 30 April 2024
- **Published in JHEP:** 25 July 2024

3.15.1.4 Quantum parity detectors: A qubit-based particle-detection scheme with meV thresholds for rare-event searches [54]

- **Authors:** Karthik Ramanathan, Brandon J. Sandoval, John E. Parker, Lalit M. Joshi, Andrew D. Beyer, Pierre M. Echternach, Serge Rosenblum, Sunil R. Golwala
- **Posted on arXiv:** 27 May 2024
- **Published in APS Open Science:** 01 May 2026

3.15.2 Dark Matter - Wave-like Dark Matter

3.15.2.1 Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor [93]

- **Authors:** Xing Rong, Mengqi Wang, Jianpei Geng, Xi Qin, Maosen Guo, Man Jiao, Yijin Xie, Pengfei Wang, Pu Huang, Fazhan Shi, Yi-Fu Cai, Chongwen Zou, Jiangfeng Du
- **Posted on arXiv:** 12 June 2017
- **Published in Nature Commun.:** 21 February 2018

3.15.2.2 Axion search with quantum nondemolition detection of magnons [94]

- **Authors:** Tomonori Ikeda, Asuka Ito, Kentaro Miuchi, Jiro Soda, Hisaya Kurashige, Yutaka Shikano
- **Posted on arXiv:** 17 February 2021
- **Published in Phys. Rev. D 105, 102004 (2022):** 15 May 2022

3.15.2.3 Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors [95]

- **Authors:** P.-H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov, V.M. Acosta
- **Posted on arXiv:** 29 December 2021
- **Published in Physical Review Research 4, 023162 (2022):** 31 May 2022

3.15.2.4 Ultimate precision limit of noise sensing and dark matter search [96]

- **Authors:** Haowei Shi, Quntao Zhuang
- **Posted on arXiv:** 29 August 2022
- **Published in npj Quantum Inf.:** 20 March 2023

3.15.2.5 Light dark matter search with nitrogen-vacancy centers in diamonds [97]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 24 February 2023
- **Published in JHEP:** 12 March 2025

3.15.2.6 Axion detection via superfluid ^{3}He ferromagnetic phase and quantum measurement techniques [98]

- **Authors:** So Chigusa, Dan Kondo, Hitoshi Murayama, Risshin Okabe, Hiroyuki Sudo
- **Posted on arXiv:** 17 September 2023
- **Published in JHEP:** 26 September 2024

3.15.2.7 Nuclear spin metrology with nitrogen vacancy center in diamond for axion dark matter detection [99]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 09 July 2024
- **Published in Phys. Rev. D 111 (2025) 075028:** 01 April 2025

3.15.2.8 Listening for new physics with quantum acoustics [100]

- **Authors:** Ryan Linehan, Tanner Trickle, Christopher R. Conner, Sohitri Ghosh, Tongyan Lin, Mukul Sholapurkar, Andrew N. Cleland
- **Posted on arXiv:** 22 October 2024
- **Published in Phys.Rev.D:** 01 December 2025

3.15.2.9 Entanglement-enhanced ac magnetometry in the presence of Markovian noise [101]

- **Authors:** Thanaporn Sichanugrist, Hajime Fukuda, Takeo Moroi, Kazunori Nakayama, So Chigusa, Norikazu Mizuochi, Masashi Hazumi, Yuichiro Matsuzaki
- **Posted on arXiv:** 28 October 2024
- **Published in Phys.Rev.A:** 07 April 2025

3.15.2.10 Probing Sub-eV Dark Photon, Scalar and Axion-like Particle Dark Matters with Transmon Qubits [102]

- **Authors:** Wei Chao, Yu Gao, Ming-Jie Jin, Xiao-Sheng Liu, Xi-Lei Sun
- **Posted on arXiv:** 30 December 2024
- **Published in Phys.Dark Univ. 50 (2025) 102171:** 15 November 2025

3.15.2.11 Piezoelectric bulk acoustic resonators for dark photon detection [103]

- **Authors:** Tanner Trickle
- **Posted on arXiv:** 09 January 2025
- **Published in Phys.Rev.D:** 01 September 2025

3.15.2.12 Background suppression in quantum sensing of dark matter via collective entangled-state projection [104]

- **Authors:** Shion Chen, Hajime Fukuda, Yutaro Iiyama, Yuya Mino, Takeo Moroi, Mikio Nakahara, Tatsumi Nitta, Thanaporn Sichanugrist
- **Posted on arXiv:** 02 October 2025
- **Published in Phys.Rev.D:** 01 March 2026

3.15.2.13 Hybrid-spin decoupling for noise-resilient DC quantum sensing [105]

- **Authors:** So Chigusa, Masashi Hazumi, Ernst David Herbschleb, Yuichiro Matsuzaki, Norikazu Mizuochi, Kazunori Nakayama
- **Posted on arXiv:** 20 November 2025

3.15.3 Crosslists

3.15.3.1 Quantum Sensing for High Energy Physics [8]

- **Authors:** Zeeshan Ahmed, Yuri Alexeev, Giorgio Apollinari, Asimina Arvanitaki, David Awschalom, Karl K. Berggren, Karl Van Bibber, Przemyslaw Bienias, Geoffrey Bodwin, Malcolm Boshier, Daniel Bowring, Davide Braga, Karen Byrum, Gustavo Cancelo, Gianpaolo Carosi, Tom Cecil, Clarence Chang, Mattia Checchin, Sergei Chekanov, Aaron Chou, Aashish Clerk, Ian Cloet, Michael Crisler, Marcel Demarteau, Ranjan Dharmapalan, Matthew Dietrich, Junjia Ding, Zelimir Djurcic, John Doyle, James Fast, Michael Fazio, Peter Fierlinger, Hal Finkel, Patrick Fox, Gerald Gabrielse, Andrei Gaponenko, Maurice Garcia-Sciveres, Andrew Geraci, Jeffrey Guest, Supratik Guha, Salman Habib, Ron Harnik, Amr Helmy, Yuekun Heng, Jason Henning, Joseph Heremans, Phay Ho, Jason Hogan, Johannes Hubmayr, David Hume, Kent Irwin, Cynthia Jenks, Nick Karonis, Raj Kettimuthu, Derek Kimball, Jonathan King, Eve Kovacs, Richard Kriske, Donna Kubik, Akito Kusaka, Benjamin Lawrie, Konrad Lehnert, Paul Lett, Jonathan Lewis, Pavel Lougovski, Larry Lurio, Xuedan Ma, Edward May, Petra Merkel, Jessica Metcalfe, Antonino Miceli, Misun Min, Sandeep Miryala, John Mitchell, Vesna Mitrovic, Holger Mueller, Sae Woo Nam, Hogan Nguyen, Howard Nicholson, Andrei Nomerotski, Mike Norman, Kevin O'Brien, Roger O'Brient, Umeshkumar Patel, Bjoern Penning, Sergey Perverzev, Nicholas Peters, Raphael Pooser, Chrystian Posada, Jimmy Proudfoot, Tenzin Rabga, Tijana Rajh, Sergio Rescia, Alexander Romanenko, Roger Rusack, Monika Schleier-Smith, Keith Schwab, Julie Segal, Ian Shipsey, Erik Shirokoff, Andrew Sonnenschein, Valerie Taylor, Robert Tschirhart, Chris Tully, David Underwood, Vladan Vuletic, Robert Wagner, Gensheng Wang, Harry Weerts, Nathan Woollett, Junqi Xie, Volodymyr Yefremenko, John Zasadzinski, Jinlong Zhang, Xufeng Zhang, Vishnu Zutshi

- **Posted on arXiv:** 29 March 2018

3.15.3.2 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [87]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

3.15.3.3 Snowmass 2021: Quantum Sensors for HEP Science – Interferometers, Mechanics, Traps, and Clocks [12]

- **Authors:** Oliver Buchmueller, Daniel Carney, Thomas Cecil, John Ellis, R.F. Garcia Ruiz, Andrew A. Geraci, David Hanneke, Jason Hogan, Nicholas R. Hutzler, Andrew Jayich, Shimon Kolkowitz, Gavin W. Morley, Holger Muller, Zachary Pagel, Christian Panda, Marianna S. Safronova
- **Posted on arXiv:** 14 March 2022

3.15.3.4 New Horizons: Scalar and Vector Ultralight Dark Matter [15]

- **Authors:** D. Antypas, A. Banerjee, C. Bartram, M. Baryakhtar, J. Betz, J.J. Bollinger, C. Boutan, D. Bowring, D. Budker, D. Carney, G. Carosi, S. Chaudhuri, S. Cheong, A. Chou, M.D. Chowdhury, R.T. Co, J.R. Crespo López-Urrutia, M. Demarteau, N. DePorzio, A.V. Derbin, T. Deshpande, M.D. Chowdhury, L. Di Luzio, A. Diaz-Morcillo, J.M. Doyle, A. Drlica-Wagner, A. Droster, N. Du, B. Döbrich, J. Eby, R. Essig, G.S. Farren, N.L. Figueroa, J.T. Fry, S. Gardner, A.A. Geraci, A. Ghalsasi, S. Ghosh, M. Giannotti, B. Gimeno, S.M. Griffin, D. Grin, D. Grin, H. Grote, J.H. Gundlach, M. Guzzetti, D. Hanneke, R. Harnik, R. Henning, V. Irsic, H. Jackson, D.F. Jackson Kimball, J. Jaeckel, M. Kagan, D. Kedar, R. Khatiwada, S. Knirck, S. Kolkowitz, T. Kovachy, S.E. Kuenstner, Z. Lasner, A.F. Leder, R. Lehnert, D.R. Leibrandt, E. Lentz, S.M. Lewis, Z. Liu, J. Manley, R.H. Maruyama, A.J. Millar, V.N. Muratova, N. Musoke, S. Nagaitsev, O. Noroozian, C.A.J. O'Hare, J.L. Ouellet, K.M.W. Pappas, E. Peik, G. Perez, A. Phipps, N.M. Rapidis, J.M. Robinson, V.H. Robles, K.K. Rogers, J. Rudolph, G. Rybka, M. Safdari, M. Safdari, M.S. Safronova, C.P. Salemi, P.O. Schmidt, T. Schumm, A. Schwartzman, J. Shu, M. Simanovskaia, J. Singh, S. Singh, M.S. Smith, W.M. Snow, Y.V. Stadnik, C. Sun, A.O. Sushkov, T.M.P. Tait, V. Takhistov, D.B. Tanner, D.J. Temples, P.G. Thirolf, J.H. Thomas, M.E. Tobar, O. Tretiak, Y.-D. Tsai, J.A. Tyson, M. Vandegar, S. Vermeulen, L. Visinelli, E. Vitagliano, Z. Wang, D.J. Wilson, L. Winslow, S. Withington, M. Wooten, J. Yang, J. Ye, B.A. Young, F. Yu, M.H. Zaheer, T. Zelevinsky, Y. Zhao, K. Zhou

- **Posted on arXiv:** 28 March 2022

3.15.3.5 Quantum Networks for High Energy Physics [16]

- **Authors:** Andrei Derevianko, Eden Figueroa, Julián Martínez-Rincón, Inder Monga, Andrei Nomerotski, Cristián H. Peña, Nicholas A. Peters, Raphael Pooser, Nageswara Rao, Anze Slosar, Panagiotis Spentzouris, Maria Spiropulu, Paul Stankus, Wenji Wu, Si Xie

- **Posted on arXiv:** 31 March 2022

3.15.3.6 Quantum Systems for Enhanced High Energy Particle Physics Detectors [115]

- **Authors:** M. Doser, E. Auffray, F.M. Brunbauer, I. Frank, H. Hillemanns, G. Orlandini, G. Kornakov

- **Published in nan:** 24 June 2022

3.15.3.7 Snowmass Computational Frontier: Topical Group Report on Quantum Computing [19]

- **Authors:** Travis S. Humble, Gabriel N. Perdue, Martin J. Savage

- **Posted on arXiv:** 14 September 2022

3.15.3.8 Quantum sensing for particle physics [108]

- **Authors:** Steven D. Bass, Michael Doser

- **Posted on arXiv:** 19 May 2023

- **Published in Nature Rev.Phys.:** 16 April 2024

3.15.3.9 Quantum Sensors for High Energy Physics [21]

- **Authors:** Aaron Chou, Kent Irwin, Reina H. Maruyama, Oliver K. Baker, Chelsea Bartram, Karl K. Berggren, Gustavo Cancelo, Daniel Carney, Clarence L. Chang, Hsiao-Mei Cho, Maurice Garcia-Sciveres, Peter W. Graham, Salman Habib, Roni Harnik, J.G.E. Harris, Scott A. Hertel, David B. Hume, Rakshya Khatiwada, Timothy L. Kovachy, Noah Kurinsky, Steve K. Lamoreaux, Konrad W. Lehnert, David R. Leibrandt, Dale Li, Ben Loer, Julián Martínez-Rincón, Lee McCuller, David C. Moore, Holger

Mueller, Cristian Pena, Raphael C. Pooser, Matt Pyle, Surjeet Rajendran, Marianna S. Safronova, David I. Schuster, Matthew D. Shaw, Maria Spiropulu, Paul Stankus, Alexander O. Sushkov, Lindley Winslow, Si Xie, Kathryn M. Zurek

- **Posted on arXiv:** 03 November 2023

3.15.3.10 Quantum measurements in fundamental physics: a user's manual [73]

- **Authors:** Jacob Beckey, Daniel Carney, Giacomo Marocco
- **Posted on arXiv:** 13 November 2023

3.15.3.11 Searches for exotic spin-dependent interactions with spin sensors [63]

- **Authors:** Min Jiang, Haowen Su, Yifan Chen, Man Jiao, Ying Huang, Yuanhong Wang, Xing Rong, Xinhua Peng, Jiangfeng Du
- **Posted on arXiv:** 04 December 2024
- **Published in Rep. Prog. Phys. 88 (2025) 016401:** 13 December 2024

3.15.3.12 Spin squeezing of macroscopic nuclear spin ensembles [64]

- **Authors:** Eric Boyers, Garry Goldstein, Alexander O. Sushkov
- **Posted on arXiv:** 19 February 2025
- **Published in Phys.Rev.D:** 01 March 2025

3.15.3.13 Symmetric Dicke States as Optimal Probes for Wave-Like Dark Matter [68]

- **Authors:** Ping He, Jing Shu, Bin Xu, Jincheng Xu
- **Posted on arXiv:** 16 December 2025

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